

Investigating Spot Market Indicators as Proxies for BESS Revenue Estimation

A Framework for Long-Term Battery Storage Revenue Evaluation

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Abstract

The global transition towards low-carbon power systems has accelerated the deployment of Battery Energy Storage Systems (BESS) to manage the variability of renewable energy generation. While BESS profitability increasingly relies on revenue stacking across multiple market segments, long-term investment evaluation is complicated by fragmented data availability across balancing and reserve products and time resolutions. This thesis develops a transparent empirical framework to test the extent to which accessible Day-Ahead spot market indicators can explain variation in long-term BESS revenue potential.

The research employs a quantitative empirical strategy using historical electricity market data from the ENTSO-E Transparency Platform for Swedish bidding zones SE3 and SE4 over 2020–2025. The methodology integrates: (i) deterministic optimisation at the *native* publication resolution of each price series to compute theoretical upper-bound revenues for Day-Ahead arbitrage and for five balancing-related revenue streams—aFRR and mFRR *activation* revenues and aFRR, mFRR, and FCR *capacity* revenues—under a six-asset virtual portfolio assumption; and (ii) regularised linear models (Ridge, Lasso, Elastic Net) together with tree ensembles (Random Forest, Gradient Boosting) that relate spot price indicators—volatility, spike frequency, and ramp rate—to the simulated daily *balancing revenue* benchmark (sum of the five streams). Capacity revenues are evaluated only in calendar years with published ENTSO-E capacity prices (aFRR capacity from 2022; mFRR and FCR capacity from 2024).

Rather than forecasting specific balancing-energy and reserve-price paths, the framework applies the tested proxy relationship to stylised Day-Ahead price scenarios (historical baseline reproduction plus three stressed scenarios) as a sensitivity exercise. Results on a 20% hold-out test set indicate that spot indicators explain only a modest fraction of variance in simulated daily *balancing revenues*, with Random Forest achieving the highest test R^2 (approximately 0.1169 in SE3 and 0.1696 in SE4). Scenario analysis indicates that the High Renewable specification yields the largest six-year cumulative uplift relative to the reproduced baseline (approximately +18.5% in SE3 and +23.2% in SE4), while balancing-revenue values in scenarios are mechanically inferred through the proxy relationship rather than directly forecasted from forward-looking balancing price paths.

Keywords: Battery Energy Storage Systems, Revenue Evaluation, Electricity Markets, Proxy-Based Relationship Testing, Day-Ahead Market, Balancing and Reserve Services, aFRR, mFRR, FCR, Virtual Portfolio

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Abbreviations

Abbreviation	Description
aFRR	Automatic Frequency Restoration Reserve
API	Application Programming Interface
BESS	Battery Energy Storage System
CET/CEST	Central European Time / Summer Time
HICP	Harmonised Index of Consumer Prices (Eurostat)
CSV	Comma-Separated Values
DST	Daylight Saving Time
ENTSO-E	European Network of Transmission System Operators for Electricity
FCR	Frequency Containment Reserve
mFRR	Manual Frequency Restoration Reserve
MTU	Market Time Unit
IRR	Internal Rate of Return
MWh / MW	Megawatt-hour / Megawatt
OLS	Ordinary Least Squares
PV	Photovoltaic
SOC	State of Charge
UTC	Coordinated Universal Time
VRES	Variable Renewable Energy Sources

1

Introduction

1.1 Background

The global transition towards low-carbon power systems has fundamentally altered the landscape of electricity generation and distribution. Driven by international climate agreements and national decarbonization targets, there has been a growing share of renewable energy generation, particularly from variable renewable energy sources (VRES) such as wind and solar photovoltaic (PV) [18, 13]. While this shift is essential for mitigating climate change, it introduces greater variability and uncertainty into electricity supply. Unlike conventional fossil-fuel power stations, which are dispatchable, VRES generation is subject to intermittent weather conditions, creating significant challenges for grid stability and load-matching capabilities [6].

To support system flexibility and reliability under these conditions, Battery Energy Storage Systems (BESS) are increasingly deployed and play an important role in modern electricity markets [28, 23]. BESS provide critical services such as energy shifting, frequency regulation, and capacity firming, allowing system operators to balance supply and demand in real-time. From a commercial perspective, the value of a battery storage project is determined by its ability to generate future revenues through market participation. These revenues are influenced by electricity price dynamics and operational decisions across different market segments, primarily the Day-Ahead (Spot) market and balancing and reserve markets [19]. In the Nordic system, key balancing and reserve products include FCR (frequency containment), aFRR (automatic frequency restoration), and mFRR (manual frequency restoration), each with distinct capacity and energy settlement components.

For investors and system planners, a key challenge lies in estimating long-term revenues in a way that is both realistic and practical, especially when market conditions are uncertain [28]. The profitability of BESS typically relies on “Revenue Stacking,” where a single asset participates in multiple market segments to maximize returns. While energy arbitrage in the Day-Ahead market is the most transparent revenue source, balancing markets generally provide much higher revenue premiums [4]. However, balancing markets exhibit strong price volatility and rely on complex marginal pricing rules, making revenues difficult to predict over long time horizons.

In practice, detailed and consistent data required for long-term revenue

assessment is not always available across all market segments. While some market data is complex or limited in historical coverage, other price signals are more transparent and widely accessible over long time horizons. This creates an opportunity to explore revenue estimation approaches that rely primarily on such accessible data sources, while still capturing the main economic drivers of battery operation.

1.2 Problem Statement

Despite the extensive literature on battery optimisation and electricity market forecasting, a structural gap remains between academically sophisticated modelling approaches and the practical needs of long-term investment evaluation. Most existing models for balancing revenues rely on direct price forecasting or complex stochastic simulations that are difficult to implement under long-term planning horizons. Furthermore, balancing market prices exhibit extreme volatility and regime-dependent behavior, making direct long-term prediction inherently uncertain and often unsuitable for transparent investment decision-making.

This modelling complexity creates a significant barrier for investors and financial stakeholders. Without a transparent method to link market dynamics to revenue outcomes, investors face high uncertainty when evaluating the economic viability of BESS projects. Existing data-driven approaches often provide improved short-term predictive accuracy but are treated as “black boxes” with limited economic interpretability. Consequently, there is limited empirical evidence on whether more interpretable and scenario-friendly Day-Ahead Spot Market indicators can meaningfully explain balancing market income under public-data constraints.

Therefore, the core problem addressed in this thesis can be formulated as follows: To what extent can a proxy-based framework using spot market indicators explain variation in long-term battery storage revenue potential, including balancing market components, under public-data constraints? Addressing this problem is critical because overconfident revenue estimation can lead to suboptimal investment decisions, either causing capital to be allocated to unviable projects or viable projects to be overlooked due to perceived risk.

1.3 Research Question and Objectives

To address the problem statement outlined above, this thesis aims to develop and evaluate a simplified empirical framework that primarily relies on Day-Ahead spot electricity prices to test whether additional balancing-market revenue benchmarks can be partly explained through observable market relationships. The study does not aim to develop a fully stochastic optimisation model nor a deep learning price forecasting architecture. Instead, it seeks to

explore a blended methodological approach that balances transparency, data efficiency, and long-term planning relevance.

The primary research question is: To what extent can Day-Ahead spot market indicators explain variation in simulated balancing-market revenue benchmarks for long-term BESS revenue evaluation under public-data constraints?

To answer this question, the thesis pursues the following specific objectives:

1. **Model multi-market upper bounds:** Estimate theoretical upper-bound revenues for Day-Ahead arbitrage and for five balancing-related streams (aFRR/mFRR activation; aFRR/mFRR/FCR capacity) using deterministic optimisation at native ENTSO-E resolution under a virtual portfolio of independent battery assets.
2. **Investigate Market Relationships:** Investigate statistical relationships between spot price indicators (e.g., volatility) and balancing revenue potential to identify viable proxy variables.
3. **Construct and test proxy relationship:** Construct and test a proxy relationship for *total* balancing revenues (sum of the five streams), calibrated on simulated historical upper-bound benchmarks, to assess whether spot indicators can support cautious scenario screening without forward-looking balancing and reserve price forecasts.
4. **Evaluate Market Scenarios:** Evaluate how different market scenarios (e.g., high volatility, high renewable penetration) influence scenario-level upper-bound revenue estimates to test the robustness of the framework.

1.4 Significance and Contributions

This research contributes to the field of energy economics and data science in several key ways. First, it addresses the identified research gap by providing a transparent and data-efficient empirical framework for testing the explanatory value of spot indicators in long-term revenue evaluation. Unlike complex stochastic models that require extensive computational resources and data, the proposed proxy-based framework is designed as a transparent first-step screening reference for investors and planners who may not have access to proprietary high-frequency data.

Second, the study provides empirical evidence on the correlation between spot market dynamics and balancing market value. By testing the use of spot price indicators as proxies, this thesis provides limited but useful evidence that observable spot dynamics contain some information about balancing market value. This finding may have implications for market design and regulatory policy, although the modest explanatory power suggests that spot indicators

should be used cautiously for investment evaluation.

Finally, the thesis offers a cautious screening framework for early-stage investment discussion. The integrated framework provides a transparent way to compare how total upper-bound revenue estimates respond under specified spot price scenarios, while recognising that the balancing component is only weakly explained by spot indicators. While the results represent theoretical upper-bound estimates, they provide a consistent benchmark for comparing market scenarios, allowing investors to apply appropriate risk discounts for operational constraints.

1.5 Delimitations and Scope

To ensure the study remains feasible within the scope of a master's thesis, several delimitations are applied. The optimization horizon is limited to historical data between 2020 and 2025, with scenario analysis extending forward-looking assumptions based on this baseline. The study focuses on the Swedish electricity market region.

Historical balancing revenues are computed only for the periods in which the relevant ENTSO-E price data are publicly available. For activation-energy products, aFRR and mFRR activation revenues are calculated from 2020 to 2025, as activated balancing energy price data are available over the full study period. For capacity products, the available public data start later: aFRR capacity revenues are calculated from 2022 onward, while mFRR and FCR capacity revenues are calculated from 2024 onward. Therefore, the five-stream balancing revenue total should be interpreted with caution, because not all balancing components are available for every year in the 2020–2025 historical sample. The scenario module does not perturb balancing and reserve price paths; stressed total revenues vary primarily through Day-Ahead arbitrage and proxy-inferred balancing components.

Broader energy system issues, such as power flow dynamics, detailed grid constraints, or policy-driven factors (e.g., specific subsidy schemes), are not within the scope of this study. Additionally, the study assumes perfect foresight regarding spot and balancing prices for the arbitrage model to establish a theoretical maximum, acknowledging that real-world operational revenues may be lower due to forecasting errors. The “virtual portfolio” approach estimates theoretical upper bounds and does not account for physical capacity conflicts in a single battery splitting capacity between markets simultaneously.

2

Literature Review

2.1 Introduction

The integration of Battery Energy Storage Systems (BESS) into modern power grids is driven by the increasing penetration of intermittent renewable energy sources. This chapter reviews the existing body of knowledge regarding BESS revenue streams, focusing on the relationship between Day-Ahead (Spot) markets and Balancing markets. It identifies the methodologies currently used for revenue evaluation and highlights the gap in simplified empirical evidence on the spot-balancing relationship under public-data constraints.

2.2 Revenue Streams for Battery Energy Storage

BESS profitability is often discussed through the concept of revenue stacking, where a storage asset earns income from several market services such as Day-Ahead arbitrage, balancing energy, and reserve capacity. However, true revenue stacking requires modelling how one physical battery allocates its limited power capacity, state of charge, and availability across competing services. This thesis does not model such co-optimised operation. Instead, the different revenue streams are estimated in parallel under a virtual portfolio assumption and then used as theoretical upper-bound benchmarks. This allows the study to compare the relative value of different market services under public-data constraints, while avoiding the claim that one single battery could realise all revenues simultaneously.

2.2.1 Arbitrage in Spot Markets

Energy arbitrage in the Day-Ahead market is the most transparent revenue source. Scholars have extensively modeled this using linear programming and deterministic optimization. Bradbury, Pratson, and Patiño-Echeverri [2] provide foundational logic for BESS arbitrage, while Jülch [14] suggest that as markets mature and price spreads narrow, pure arbitrage alone often fails to meet Internal Rate of Return (IRR) targets without additional service stacking.

2.2.2 Balancing and Reserve Services

Balancing markets such as FCR and aFRR generally provide much higher revenue premiums than pure energy arbitrage in the spot market. Mohamed et al. [19] highlight that balancing services have become a main revenue driver for BESS business cases in Europe, often accounting for a large share of total

annual income. However, as discussed by Burger et al. [4], these markets exhibit strong price volatility and rely on complex marginal pricing rules. As a result, balancing revenues are difficult to predict over long time horizons, which creates significant uncertainty for long-term investment decisions and strategic asset planning. European transparency rules require publication of contracted reserve prices and activated balancing energy prices; however, data availability and temporal resolution vary by product and bidding zone. This motivates a product-by-product upper-bound decomposition rather than a single aggregated balancing revenue series.

2.2.3 Methodologies for Revenue Estimation

The literature presents several approaches to estimating potential BESS income:

Deterministic Optimization

Deterministic models assume perfect foresight of prices. Braeuer et al. [3] argue that while useful for calculating “theoretical maximums,” these models often overestimate real-world revenues by 20%–30% due to the lack of perfect price knowledge. Earlier foundational work by Sioshansi et al. [25] modeled storage arbitrage in competitive electricity markets under perfect price information, establishing the baseline logic of profit-maximizing dispatch. Similarly, Denholm and Margolis [8] evaluated long-term storage value under different renewable penetration scenarios and found that deterministic dispatch tends to produce upper-bound revenue estimates. Xu et al. [27] further show that perfect-foresight optimisation leads to systematically higher revenue outcomes compared to rolling-horizon strategies that incorporate forecast errors. These findings suggest that while deterministic models are transparent and computationally efficient, they may provide optimistic revenue projections when applied to long-term investment evaluation.

Stochastic and Probabilistic Modeling

Stochastic approaches explicitly account for price uncertainty by simulating multiple possible price outcomes. For example, Hameed, Træholt, and Hashemi [10] propose a scenario-based stochastic planning model for BESS participation across energy and regulation markets, capturing uncertain market prices as well as penalties and degradation costs. Conejo, Carrión, and Morales [5] present stochastic programming frameworks using scenario trees to represent electricity price uncertainty in market participation problems. Morales et al. [20] discuss probabilistic approaches that model full price distributions rather than point forecasts, improving risk representation. Zugno et al. [31] incorporate uncertainty in both energy and reserve markets within a stochastic optimisation setting for storage operation. While such stochastic approaches provide a more realistic representation of uncertainty, they are

computationally intensive and highly sensitive to scenario construction and input assumptions. Their practical implementation for long-term revenue estimation remains challenging when high-resolution data is unavailable.

Data-Driven Approaches

Recent shifts toward deep learning methods, such as the application of LSTM and CNN models for electricity market price forecasting, have demonstrated improved short-term predictive accuracy compared to traditional approaches [15]. Lago et al. [17] provide a systematic review of deep learning techniques for electricity price forecasting and conclude that neural network architectures outperform classical statistical models in short-horizon predictions. Zhang, Wang, and Zhang [29] combine convolutional and recurrent neural network structures to capture both spatial and temporal dependencies in electricity price series. Weron [26] critically review statistical and machine learning forecasting models in electricity markets and emphasize that long-term forecasting remains inherently difficult due to structural market changes and regulatory shifts. However, these models are often treated as “black boxes” and provide limited economic interpretability, which reduces their suitability for long-term strategic planning and investment decision-making by financial stakeholders.

2.3 The Relationship Between Spot and Balancing Prices

A critical area of research is the correlation between these two markets. Pape, Hagemann, and Weber [21] were among the first to explore how scarcity in the spot market correlates with high balancing costs in the German frequency restoration reserve market. More recently, research from the FfE Institute [9] suggests that spot market price signals can serve as indicators for balancing market trends, providing a basis for testing whether balancing revenue patterns can be partly explained using more accessible datasets. While these relationships do not imply a one-to-one mapping between spot and balancing prices, they provide a useful basis for proxy-based relationship testing.

Further empirical evidence supports the use of spot market indicators as proxies for balancing market conditions. For instance, Hirth and Ziegenhagen [12] demonstrate that intraday price volatility in the Day-Ahead market is associated with system imbalance costs in Central European markets. Similarly, Knaut and Paschmann [16] find that extreme price events in the spot market are statistically associated with increased activation of balancing reserves, suggesting that observable spot dynamics may contain limited explanatory information about balancing market value.

2.4 Identification of Research Gaps

Despite the wealth of literature on BESS optimization, two major gaps persist:

- **Forecasting Complexity:** Many balancing studies rely on product-specific high-frequency series and proprietary dispatch; public feeds differ by zone, product, and year, complicating long panels.
- **Limited Evidence on Simplified Proxies:** There is limited empirical evidence on whether Day-Ahead Spot Market indicators can meaningfully explain balancing market income under public-data constraints.

3

Methodology

3.1 Research Method and Strategy

This study adopts a quantitative empirical research strategy to investigate the potential revenue generation of Battery Energy Storage Systems (BESS) in electricity markets. The research focuses on analysing market price data and modelling operational behaviour of battery systems under different market conditions [7, 22].

3.1.1 Alternative Research Strategies

Several alternative strategies could be used to address the research question, but were not selected as the primary design for this thesis:

- **Qualitative case study / interviews:** In-depth stakeholder interviews (e.g., traders, TSOs, storage operators) could provide rich insight into bidding strategies, risk constraints, and contractual realities. However, these approaches do not directly produce a reproducible, data-driven estimate of revenue potential and are sensitive to respondent access and confidentiality constraints.
- **Mixed-methods (qualitative + quantitative):** Combining interviews with modelling could triangulate operational assumptions (e.g., dispatch restrictions, bidding rules). This can be powerful, but is beyond the feasible scope of this study and would require additional primary data collection and ethics approval workflows.
- **Stochastic optimisation / agent-based simulation:** A stochastic co-optimisation model with forecast errors and multi-market bidding would better represent real trading conditions, but it requires detailed probabilistic inputs (e.g., forecast distributions, imbalance settlement rules, activation volumes) that are not consistently available from public data and would substantially increase model complexity.

Given the aim of producing a transparent, replicable, and data-efficient framework for investment screening under limited data availability, the thesis adopts a quantitative empirical strategy based on publicly available market data.

3.1.2 Ethical Considerations

This thesis uses publicly available market datasets (ENTSO-E Transparency Platform and Eurostat HICP) and does not process personal data. Ethical considerations therefore relate primarily to responsible research practice: (i) transparency of modelling assumptions (e.g., perfect foresight, virtual portfolio upper bounds), (ii) reproducibility (code-based data retrieval and logged exclusions), and (iii) cautious interpretation to avoid overstating results as operationally achievable revenues for a single physical asset.

3.1.3 Research Approach

A quantitative approach is appropriate because the research question concerns measurable economic outcomes. The strategy combines three complementary analytical components:

1. **Deterministic Optimisation:** Used to model battery arbitrage behaviour in the Day-Ahead (Spot) market and in balancing and reserve markets under perfect foresight assumptions.
2. **Statistical and Machine-Learning Proxy Modelling:** Investigates the relationship between spot market indicators and *total balancing* revenue potential using Ridge, Lasso, Elastic Net, Random Forest, and Gradient Boosting.
3. **Scenario Analysis:** Evaluates how stressed Day-Ahead market conditions (e.g., volatility, renewable penetration) influence upper-bound revenue estimates, with balancing components inferred through the tested proxy relationship as a sensitivity exercise.

3.1.4 Research Design

This research employs a quantitative experimental design using historical market data as a testbed. The design involves:

- **Data Acquisition:** Collecting price data via programmatic interfaces (ENTSO-E API).
- **Model Implementation:** Simulating battery dispatch and constructing upper-bound revenue benchmarks.
- **Scenario Testing:** Applying structural adjustments to simulate future market conditions.

3.2 Data Collection and Description

3.2.1 Data Sources and Retrieval

The primary data consists of publicly available market data from the **ENTSO-E Transparency Platform**.

- **Day-Ahead spot prices:** Day-Ahead prices (A44, contract A01) for Swedish bidding zones SE3 and SE4, retained at native ENTSO-E resolution.
- **Balancing and reserve prices:** Activated balancing energy prices (A84) for aFRR and mFRR; contracted reserve prices (A81) for aFRR, mFRR, and FCR capacity, each retained at native resolution.
- **Time horizon:** 1 January 2020 (inclusive) to 1 January 2026 (exclusive), covering deliveries through 31 December 2025.

3.2.2 Products, Documents, and Sample Windows

The balancing-revenue analysis covers three reserve products—FCR, aFRR, and mFRR—and two settlement layers where applicable: *activation* (energy) and *capacity* (reserved power). Table 3.1 summarises ENTSO-E document types, key filters, native resolution, and the calendar window used in this thesis.

Table 3.1. ENTSO-E data mapping and effective sample windows (SE3/SE4)

Revenue stream	Doc.	Key filters	Native res.	From
Day-Ahead arbitrage	A44	contract A01	1 h; 15 min [†]	2020
aFRR activation	A84	A96, process A16	1 h; 15 min [‡]	2020
mFRR activation	A84	A97, process A16	1 h; 15 min [‡]	2020
aFRR capacity	A81	process A51, type A01	15 min	2022
mFRR capacity	A81	process A47, type A01	15 min	2024
FCR capacity	A81	process A52, type A01	15 min	2024

[†]15-minute Day-Ahead from 1 October 2025.

[‡]15-minute activation prices from approximately June 2025.

Activated volumes (A83) are unavailable for Sweden via the public API.

Day-Ahead and activation prices are optimised at their native step length within each delivery day ($T_d \in \{24, 96\}$ depending on the date). Capacity prices are optimised at 15-minute resolution where published. No capacity revenues are computed before the product-specific start year in Table 3.1.

3.2.3 Data Preprocessing

Time Alignment: Timestamps converted from UTC to the Europe/Stockholm timezone (CET/CEST), with DST transitions handled (duplicate hours removed, keeping the last observation).

Missing Values: Short gaps are interpolated in time with a limit expressed in *native steps* (equivalent to ≤ 2 hours for hourly series and ≤ 8 steps for 15-minute series), using `limit_direction='both'`; longer gaps remain missing.

Resolution: No uniform hourly aggregation is applied prior to optimisation; each product is dispatched at the finest published resolution for that day.

Inflation Adjustment: Nominal prices adjusted using the Swedish all-items Harmonised Index of Consumer Prices (HICP) from Eurostat (dataset prc_hicp_midx, CP00, 2015=100).

3.3 Spot Market Arbitrage Model

3.3.1 Optimisation Formulation

The objective is to maximise daily arbitrage profit (TR_d):

$$\max \sum_{t=1}^{T_d} (p_t \cdot \text{discharge}_t \cdot \Delta t - p_t \cdot \text{charge}_t \cdot \Delta t) - c_{\text{deg}} \sum_{t=1}^{T_d} (\text{charge}_t + \text{discharge}_t) \cdot \Delta t \quad (3.1)$$

Subject to operational constraints:

- **SOC Dynamics:** $SOC_t = SOC_{t-1} + \eta \cdot \text{charge}_t \cdot \Delta t - \frac{\text{discharge}_t \cdot \Delta t}{\eta}$
- **Power Limits:** $0 \leq \text{charge}_t, \text{discharge}_t \leq P_{\max}$
- **Mutual Exclusivity:** Charge and discharge are mutually exclusive in each native time step (MTU) (implemented via a binary on/off decision to prevent simultaneous charging and discharging).
- **Capacity Limits:** $SOC_{\min} \leq SOC_t \leq SOC_{\max}$
- **Cycle Limit:** $\sum_{t=1}^{T_d} (\text{charge}_t + \text{discharge}_t) \leq 2 \cdot C_{\max} \cdot E_{\max}$

For each delivery day d , T_d denotes the number of native Day-Ahead intervals (24 before 1 October 2025 and 96 thereafter for Swedish zones in the ENTSO-E feed used in this study). Power variables remain in MW; energy throughput in the SOC constraint is consistent with the interval length Δt .

3.3.2 Parameter Specification and Initial Conditions

To ensure the model reflects realistic operational characteristics of utility-scale lithium-ion battery systems, the technical parameters are fixed based on representative values found in recent literature [3, 23]. The specific values applied throughout the analysis are summarised in Table 3.2.

Table 3.2. Battery Technical Parameters and Operational Constraints

Parameter	Symbol	Value
Energy Capacity	$E_{\max}$	100 MWh
Power Capacity	$P_{\max}$	25 MW
C-rate	–	0.25 h^{-1} (4-hour system)
Round-trip Efficiency	η	0.90
Minimum SOC	$SOC_{\min}$	10 MWh ($0.1 \cdot E_{\max}$)
Maximum SOC	$SOC_{\max}$	100 MWh ($1.0 \cdot E_{\max}$)
Max Daily Cycles*	$C_{\max}$	2 equivalent full cycles
Degradation Cost	c_{deg}	20 EUR/MWh

**This constraint is applied as a modeling simplification to prevent excessive intra-day arbitrage in the deterministic optimization framework, rather than representing a strict technical limitation of battery hardware.*

Initial State of Charge. For the first day of the simulation horizon ($d = 1$), the initial state of charge is set to 50% of the energy capacity ($SOC_0 = 50 \text{ MWh}$). For all subsequent days ($d > 1$), the optimisation employs a rolling horizon approach where the initial SOC of day d is constrained to equal the terminal SOC of day $d - 1$. This inter-temporal coupling ensures energy continuity across the multi-year simulation period.

Degradation Modelling. Battery degradation is approximated as a linear cost per unit of energy throughput (c_{deg}), rather than a complex cycle-counting function. This simplification is computationally efficient for long-term horizon optimisation and provides a reasonable first-order approximation of operational expenditure (OPEX) related to battery wear [27].

3.4 Balancing Revenue Models

3.4.1 Rationale

In this thesis, the term **balancing revenues** refers to the simulated upper-bound revenues from the five balancing-related streams modelled in the study: aFRR and mFRR activation revenues, and aFRR, mFRR, and FCR capacity revenues. Although these products may also be described as ancillary or reserve services in broader electricity-market terminology, the thesis uses the term balancing revenues for consistency with the proposal, research question, and the empirical revenue variable BR_d .

In balancing and reserve markets, battery storage systems provide flexibility services to support real-time system balance. In this thesis, balancing revenues are estimated independently from spot arbitrage revenues. The spot and balancing models are treated as separate application cases, equivalent to independently allocated battery systems participating in different markets. This “virtual portfolio” approach provides a transparent benchmark for the

theoretical upper-bound value of revenue stacking [3, 24].

Six independent optimisation problems are solved: one Day-Ahead arbitrage stream and five balancing-related streams (two activation, three capacity). Revenues are summed only after optimisation: $BR_d = \sum_k BR_{d,k}$.

3.4.2 Activation-Energy Optimisation (aFRR and mFRR)

For product $k \in \{\text{aFRR}, \text{mFRR}\}$, let $\pi_{t,k}^\uparrow$ and $\pi_{t,k}^\downarrow$ denote activation prices (EUR/MWh) at native step t within delivery day d . The virtual battery maximises

$$\max \sum_{t=1}^{T_d} \left(\pi_{t,k}^\uparrow p_{t,k}^{\text{dis}} \cdot \Delta t - \pi_{t,k}^\downarrow p_{t,k}^{\text{ch}} \cdot \Delta t \right) - c_{\text{deg}} \sum_{t=1}^{T_d} \left(p_{t,k}^{\text{ch}} + p_{t,k}^{\text{dis}} \right) \cdot \Delta t \quad (3.2)$$

subject to the same SOC, power, mutual-exclusivity, and daily cycle limits as the spot model (Section 3.3.2), with Δt matched to the native MTU. This formulation is a perfect-foresight upper bound on energy settlement; it does not model automatic frequency response duty cycles.

3.4.3 Capacity-Revenue Optimisation (aFRR, mFRR, FCR)

Capacity markets compensate reserved power (MW). For aFRR and mFRR, let $R_{t,k}^\uparrow$ and $R_{t,k}^\downarrow$ denote upward and downward reserved capacity. The virtual capacity battery maximises

$$\max \sum_{t=1}^{T_d} \left(\pi_{t,k}^{\text{cap},\uparrow} R_{t,k}^\uparrow + \pi_{t,k}^{\text{cap},\downarrow} R_{t,k}^\downarrow \right), \quad 0 \leq R_{t,k}^\uparrow, R_{t,k}^\downarrow \leq P_{\text{max}}. \quad (3.3)$$

Under the virtual-portfolio convention, upward and downward capacity can each reach P_{max} independently (no shared-power constraint), which strengthens the upper bound.

For FCR, a single symmetric capacity price $\pi_t^{\text{cap},\text{sym}}$ applies:

$$\max \sum_{t=1}^{T_d} \pi_t^{\text{cap},\text{sym}} R_t^{\text{sym}}, \quad 0 \leq R_t^{\text{sym}} \leq P_{\text{max}}. \quad (3.4)$$

No state-of-charge dynamics are modelled for capacity streams. Without published activated volumes (A83), Equations (3.3)–(3.4) imply that capacity can be reserved whenever prices are positive—a deliberate theoretical ceiling.

3.5 Construction of Balancing Revenue Proxy

In the previous section, spot market arbitrage revenues were estimated using deterministic optimisation based on Day-Ahead electricity prices and battery operational constraints. This optimisation captures the energy trading value of the battery.

However, spot arbitrage revenues alone may not represent the full economic

value of battery storage, as balancing and reserve markets can contribute a significant share of total revenues. These revenues are often associated with system stress conditions such as high volatility, scarcity events, and large price movements, which are reflected in observable spot price behaviour.

To capture this relationship, daily spot price indicators—volatility, spike frequency, and ramp rate—are used as explanatory variables in a regularised regression model where simulated *total balancing* revenues serve as the dependent variable. The tested proxy relationship can then be used to examine how balancing-revenue estimates respond under different spot market scenarios, without forecasting product-specific balancing and reserve price paths.

3.5.1 Rationale

Instead of modelling balancing and reserve market outcomes forward-looking, this study examines whether balancing revenue benchmarks can be indirectly related to indicators derived from observable spot price behaviour.

In particular, several spot price characteristics are constructed to capture underlying electricity market conditions, including price volatility, frequency of extreme price events, and large intraday price changes. These indicators reflect market stress conditions that are often associated with higher balancing value.

Using these spot-based indicators, the study develops proxy variables to test whether *total* balancing revenue benchmarks can be partly explained by spot-market conditions. The relationship between these indicators and historical simulated balancing upper bounds is subsequently examined using regularised regression analysis, allowing the analysis to assess the degree to which BR_d is statistically associated with observable spot market dynamics.

3.5.2 Regularised Regression Specification

Daily balancing revenue is denoted as BR_d , where d indexes days. In this thesis, BR_d is defined as the sum of five simulated balancing-related streams (aFRR/mFRR activation; aFRR/mFRR/FCR capacity) within each product's valid sample window (Table 3.1).

Three spot market indicators are constructed as explanatory variables:

1. **Price Volatility** (Vol_d): Standard deviation of the *hourly-resampled* Day-Ahead prices within day d (24 values), to ensure indicator comparability across 2020–2025 despite a late-sample switch to 15-minute publication.
2. **Spike Frequency** ($Spike_d$): Count of hours where $p_{d,t} > P_{95}$, where P_{95} is the 95th percentile of spot prices over the pooled sample period (both SE3 and SE4).
3. **Maximum Absolute Ramp** ($Ramp_d$): $\max_{t=2}^{24} |p_{d,t} - p_{d,t-1}|$ computed on the same hourly-resampled Day-Ahead series.

An additional candidate indicator, the intraday spread (max–min), was excluded from estimation due to near-perfect collinearity with volatility. After excluding spread, residual correlation between volatility and ramp rate remained; therefore, regularised regression methods are applied.

The balancing revenue proxy model is specified as:

$$BR_d = \beta_0 + \beta_1 Vol_d + \beta_2 Spike_d + \beta_3 Ramp_d + \varepsilon_d \quad (3.5)$$

Three regularised regression methods are applied:

Ridge Regression (L2 penalty).

$$\min_{\beta} \sum_{d=1}^N \left(BR_d - \hat{BR}_d \right)^2 + \lambda \sum_{j=1}^3 \beta_j^2 \quad (3.6)$$

Lasso Regression (L1 penalty).

$$\min_{\beta} \sum_{d=1}^N \left(BR_d - \hat{BR}_d \right)^2 + \lambda \sum_{j=1}^3 |\beta_j| \quad (3.7)$$

Elastic Net Regression (combined L1/L2 penalty). Elastic Net combines coefficient shrinkage and variable selection by mixing Ridge and Lasso penalties:

$$\min_{\beta} \sum_{d=1}^N \left(BR_d - \hat{BR}_d \right)^2 + \lambda \left[\alpha \sum_{j=1}^3 |\beta_j| + (1 - \alpha) \sum_{j=1}^3 \beta_j^2 \right] \quad (3.8)$$

where $\lambda \geq 0$ controls the overall penalty strength and $\alpha \in [0, 1]$ controls the mix between Lasso and Ridge. When $\alpha = 1$, Elastic Net reduces to Lasso; when $\alpha = 0$, it reduces to Ridge [11, 30].

Model Validation. The dataset is randomly split into training (80%) and test (20%) sets with a fixed seed for reproducibility. Features are standardised (zero mean, unit variance) using parameters fitted on the training set only. Model performance is assessed using:

- **Test set R^2 :** Proportion of variance in balancing revenues explained by the model on unseen data
- **Mean Absolute Error (MAE):** Average absolute error between proxy-fitted and simulated balancing revenue benchmarks in EUR/day
- **Root Mean Squared Error (RMSE):** Penalises larger deviations between proxy-fitted and simulated balancing revenue benchmarks more heavily

Note on Validation Strategy. While random train-test splitting is commonly used for cross-sectional data, electricity price series exhibit temporal dependence and autocorrelation. Random splitting may therefore underestimate

out-of-sample error by allowing information from adjacent time periods to leak between training and test sets. For this thesis, given the focus on establishing a proof-of-concept proxy framework with a static historical dataset (2020–2025), random splitting provides an initial assessment of model generalisability. However, future extensions should employ time-series cross-validation (e.g., rolling window or expanding window approaches) to better respect temporal ordering and provide more realistic estimates of out-of-sample performance under sequential forecasting conditions.

Nonlinear benchmark models. In addition to regularised linear models, two nonlinear benchmark models are estimated: Random Forest and Gradient Boosting. These models test whether nonlinearities or interaction effects between spot indicators improve out-of-sample explanatory performance. They are used as benchmarks for assessing whether the linear specification misses major nonlinear patterns.

Interpretation and Application. The tested proxy relationship (Equation 3.5) is then applied to stylised spot price scenarios as a sensitivity exercise, in order to examine how proxy-inferred balancing components respond under alternative market structures. Because the relationship requires only spot price indicators as inputs—not forward-looking balancing-energy and reserve-price paths—it supports cautious screening-level comparison even when balancing market forecasts are unavailable.

3.6 Scenario Construction

To evaluate revenue performance under alternative market structures, the historical baseline price path (2020–2025) is reproduced and three stressed Day-Ahead spot price scenarios are constructed. This scenario module is a deterministic sensitivity exercise, not a forecast. Scenario construction applies only to Day-Ahead prices; balancing and reserve price series are not stressed forward, and scenario balancing-revenue values are mechanically inferred through the tested proxy relationship applied to scenario spot indicators.

The scenarios are:

1. **High Volatility:** Represents a system with higher renewable penetration but limited flexibility resources
2. **Low Volatility:** Represents a system with significant flexibility deployment alongside renewable expansion
3. **High Renewable Penetration:** Represents a system where solar and wind become dominant marginal generation sources

Price Decomposition. All scenarios are constructed by decomposing the baseline hourly Day-Ahead price series p_t into:

$$p_t = \mu_t + \varepsilon_t \quad (3.9)$$

where μ_t is the deterministic structural component (month-hour sample mean) and ε_t is the stochastic residual.

Scenario Parameters. Table 3.3 summarises the scenario construction parameters.

Table 3.3. Summary of Scenario Construction Parameters

Scenario	Mechanism	Parameter Value
High Volatility	Stochastic scaling	$k_{HV} = 1.5$
Low Volatility	Stochastic scaling	$k_{LV} = 0.6$
High Renewables	Structural shift + scaling	$\Delta_{mid} = -25, \Delta_{eve} = +35$

High Volatility Scenario. The high volatility scenario amplifies stochastic deviations:

$$p_t^{HV} = \mu_t + 1.5 \cdot \varepsilon_t \quad (3.10)$$

This corresponds to a 50% increase in stochastic deviation from structural price levels, representing a moderate stress scenario rather than an extreme crisis regime.

Low Volatility Scenario. The low volatility scenario dampens stochastic deviations:

$$p_t^{LV} = \mu_t + 0.6 \cdot \varepsilon_t \quad (3.11)$$

This implies a 40% reduction in stochastic price deviations, representing a stabilized market environment.

High Renewable Penetration Scenario. Unlike the volatility scenarios, the high renewable scenario explicitly adjusts the intraday price structure to reflect structural changes in generation mix:

$$p_t^{RE} = \begin{cases} \mu_t - 25 + 1.2 \cdot \varepsilon_t, & t \in [11, 15] \text{ (midday)} \\ \mu_t + 35 + 1.2 \cdot \varepsilon_t, & t \in [18, 22] \text{ (evening)} \\ \mu_t + 1.2 \cdot \varepsilon_t, & \text{otherwise} \end{cases} \quad (3.12)$$

This specification captures:

- Lower midday prices due to excess solar generation (-25 EUR/MWh)
- Higher evening peak prices due to reduced solar output ($+35$ EUR/MWh)
- Moderately higher volatility reflecting renewable intermittency ($k_{RE} = 1.2$)

Price Floor. To avoid economically implausible outcomes, all scenario prices are floored at -150 EUR/MWh. The price floor of -150 EUR/MWh is set as a conservative scenario parameter, well within Nord Pool’s technical limit of -500 EUR/MWh, reflecting realistic negative price observations in European markets.

Application to Proxy Model. For each scenario, daily spot indicators (volatility, spike frequency, ramp rate) are recomputed using the same rules as Section 3.5.2, with spike counts defined relative to the historical pooled 95th percentile to ensure cross-scenario comparability. For scenario application, the calibrated Lasso proxy relationship is refit on the full historical panel (with 5-fold cross-validation for λ selection), rather than the 80/20 train/test split used for hold-out evaluation. The resulting relationship is then applied to infer screening-level balancing components, which are aggregated with spot arbitrage profits from the deterministic optimisation to obtain scenario-level upper-bound revenue estimates.

This pipeline is a deterministic sensitivity exercise, not a forecast: it evaluates how scenario-level upper-bound revenue estimates respond to stylised structural changes in the spot price process.

3.7 Integration of Spot Arbitrage and Balancing Revenues

This section integrates the deterministic spot arbitrage optimisation model with proxy-inferred balancing revenue components for scenario sensitivity analysis. Under the virtual battery assumption, the total revenue potential is defined as the sum of the independent revenue streams.

Daily total revenue is defined as:

$$TR_d = AR_d + BR_d \quad (3.13)$$

where:

- AR_d is the optimised daily spot arbitrage revenue
- BR_d is the proxy-inferred daily *total balancing* revenue component from the tested proxy relationship (Equation 3.5)

This integrated framework enables comparison of scenario-level upper-bound revenue estimates under specified spot price scenarios, even when forward-looking balancing market data are unavailable. It represents the theoretical revenue capacity of an asset portfolio capable of accessing both market segments independently; this aggregate value should be interpreted as an upper-bound benchmark for cautious screening, rather than a guaranteed operational cash flow for a single physical unit.

3.7.1 Virtual Portfolio Assumption

The integration approach adopts a “virtual portfolio” framework, wherein spot arbitrage and balancing market participation are modelled as separate, independent revenue streams. This methodological choice is deliberate and serves several purposes:

1. **Transparency:** By isolating each revenue stream, the framework provides clear visibility into the contribution of each market segment to total revenue potential.
2. **Upper-bound estimation:** The virtual portfolio approach yields theoretical maximum revenues, establishing a ceiling against which real-world operational constraints can be evaluated.
3. **Modularity:** The separation allows independent calibration and validation of each component model before integration.

However, a single physical battery cannot necessarily realise all revenue streams simultaneously. In practice, the battery’s state of charge and power capacity must be allocated across competing services, and market rules may restrict simultaneous participation. For this reason, the revenue estimates in this thesis should be interpreted as upper-bound revenue potential under a virtual portfolio framework. The thesis does not attempt to estimate a feasible stacked revenue for a single physical battery, as doing so would require detailed multi-market dispatch data and operational constraints beyond the available public dataset.

3.7.2 Application to Scenario Analysis

The integrated framework is applied to each scenario constructed in Section 3.6. For each scenario, Day-Ahead arbitrage revenues are re-optimised using the scenario-specific spot price series. Spot indicators are then recomputed from the same scenario prices and passed into the tested proxy relationship to infer balancing-revenue components. Finally, the two components are aggregated to obtain scenario-level upper-bound revenue estimates for comparison.

Because the integration follows the virtual portfolio assumption, scenario totals are interpreted as upper-bound benchmarks. The main purpose of the scenario analysis is therefore to compare how different Day-Ahead price structures affect estimated revenue potential, rather than to forecast the exact cash flow of a single physical battery.

3.8 Methodological Limitations

Several limitations must be acknowledged:

- **Perfect Foresight:** The deterministic optimisation models assume known prices within each delivery day at native resolution (T_d intervals), yielding theoretical upper-bound revenues. Real-world trading with forecast errors would capture a lower proportion of these profits [3, 27].
- **Virtual Portfolio:** Revenues from spot arbitrage and balancing markets are estimated independently and aggregated post-hoc. This creates a theoretical upper-bound benchmark rather than a feasible stacked revenue estimate for a single physical battery. A real battery would face state-of-charge competition, power capacity conflicts, bidding restrictions, and market participation constraints [24].
- **Temporal resolution heterogeneity:** Day-Ahead, activation, and capacity series publish at different native resolutions that change over the sample (Table 3.1). Results therefore mix hourly and 15-minute upper bounds across years rather than enforcing a single MTU.
- **Capacity sample truncation:** aFRR capacity revenues begin in 2022; mFRR and FCR capacity revenues begin in 2024. Combined five-stream totals are not comparable across the full 2020–2025 window without interpretation.
- **No volume-weighting:** Without activated volumes (A83), activation revenues use price–dispatch upper bounds; capacity revenues assume full $P_{\max}$ availability whenever remunerated.
- **Proxy Relationship Stability:** The tested relationship between spot indicators and simulated balancing revenue benchmarks assumes structural stability over time. Regulatory changes, market design reforms, or shifts in generation mix could alter this relationship, limiting out-of-sample applicability.
- **Parameter Point Estimates:** Technical parameters (efficiency, degradation cost, cycle limits) are fixed at representative values. Sensitivity analysis could quantify revenue uncertainty under technical variation, but is beyond the scope of this study.

Despite these limitations, the methodology prioritises transparency and long-term usability over short-term forecasting precision. The theoretical upper-bound estimates provide a consistent benchmark for comparing market scenarios, while investors are advised to apply appropriate risk discounts for operational constraints.

4

Results

This chapter presents the empirical outcomes of the proposed revenue estimation framework. The analysis proceeds in four parts: (1) Day-Ahead arbitrage, (2) balancing revenues (stacked aFRR/mFRR activation and aFRR/mFRR/FCR capacity streams), (3) proxy modelling of daily balancing revenues, and (4) scenario analysis. All monetary values are expressed in real 2025 EUR unless otherwise noted.

4.1 Spot Market Arbitrage Revenues

4.1.1 Price Dynamics and Summary Statistics

Table 4.1 summarises daily profit statistics. SE4 delivers higher mean profits than SE3, reflecting stronger coupling with continental markets. The large gap between mean and median (e.g., SE3: EUR 2,985 vs. EUR 39) indicates a strong right-skewed distribution.

Table 4.1. Spot arbitrage: summary statistics (2020–2025). *Combined* is calculated from the daily sum of SE3 and SE4 revenues; therefore, the combined median and standard deviation are computed from the combined daily series and do not necessarily equal the sum of zone-level summary statistics. The combined series does not account for physical capacity conflicts in a single asset.

Metric	SE3 (EUR)	SE4 (EUR)	Combined* (EUR)
Mean daily profit	2,985	3,880	6,865
Std. dev. daily profit	6,418	7,062	12,819
Median daily profit	39	899	1,599
Total Revenue	6.55M	8.51M	15.06M

4.1.2 Empirical results and visual evidence

This subsection presents the main spot-arbitrage results, supported by figures that highlight price dynamics, dispatch behaviour, and revenue patterns.

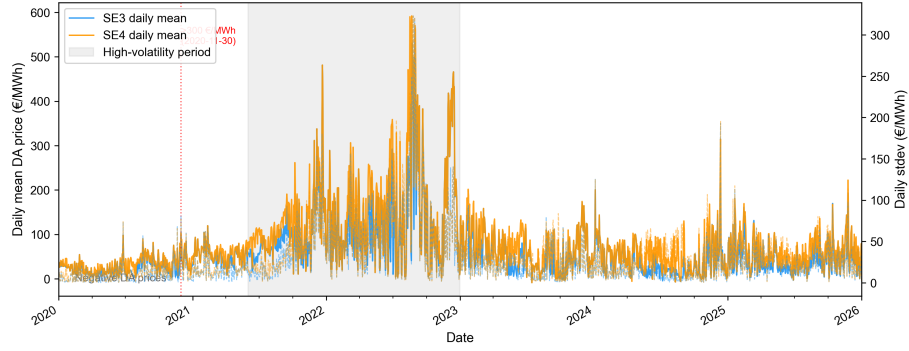


Fig. 4.1. Spot market price timeseries

Figure 4.1 shows the evolution of Day-Ahead spot prices for SE3 and SE4. Both zones exhibit a pronounced regime shift around 2021–2022, with elevated mean prices and intraday volatility during the energy crisis period. SE4 consistently shows higher volatility than SE3, reflecting its stronger coupling with continental European markets via interconnectors. Negative prices and extreme spikes (>300 EUR/MWh) are retained as economically meaningful extremes for battery arbitrage.

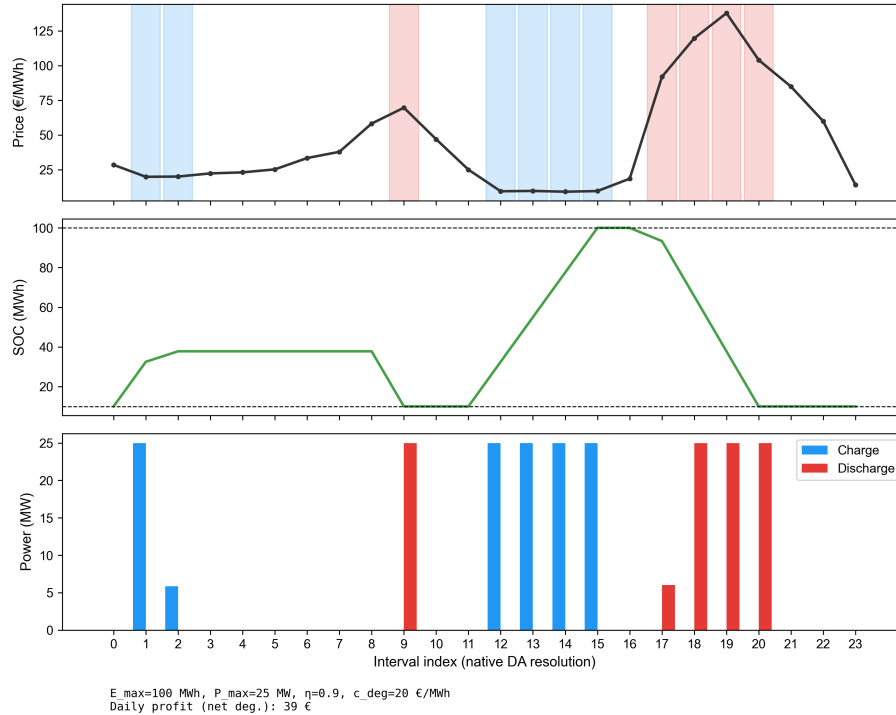


Fig. 4.2. Example dispatch strategy

Figure 4.2 illustrates the optimal dispatch strategy on a representative day. The battery charges during low-price hours (e.g., overnight and midday) and discharges during high-price peaks (morning and evening), respecting power limits ($P_{\max} = 25$ MW) and SOC bounds (10–100 MWh). The net daily profit reported for the example day (after degradation cost) exemplifies the “fat-tail” nature of arbitrage returns: most days yield modest profits, while extreme price

spreads generate occasional large gains.

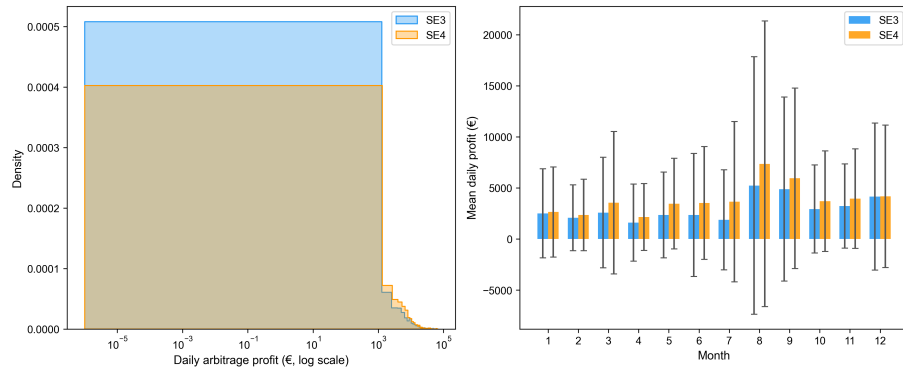


Fig. 4.3. Revenue distribution

Figure 4.3 (left) confirms the right-skewed distribution of daily arbitrage profits: the bulk of observations cluster near zero, with a long right tail extending to above EUR 60,000/day. This pattern is consistent with electricity market microstructure, where value is concentrated in scarcity events. The right panel shows that average daily arbitrage profits vary across calendar months, but the large error bars indicate substantial dispersion within each month. Therefore, the seasonal pattern should be interpreted cautiously rather than as a stable winter-only effect.

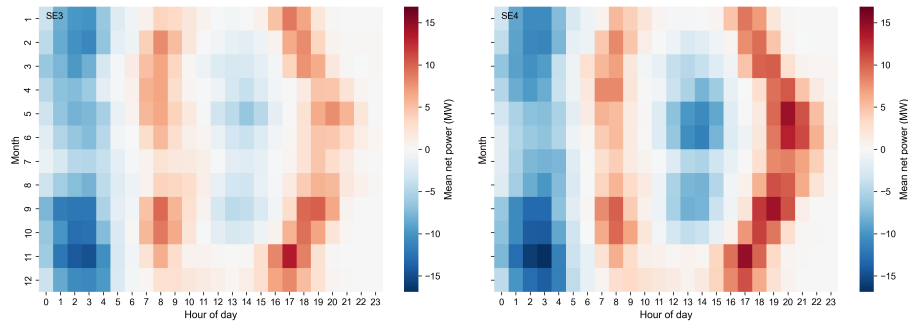


Fig. 4.4. Dispatch heatmap

Figure 4.4 visualises the optimal hourly dispatch pattern across the full sample. Charging concentrates in recurring low-price periods, while discharging clusters in high-price peaks, consistent with the arbitrage logic.

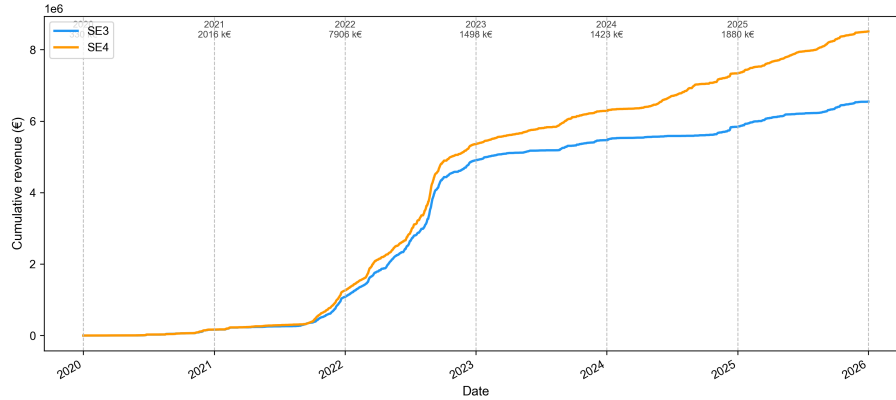


Fig. 4.5. Cumulative revenue

Figure 4.5 visualises the path-dependence of cumulative arbitrage revenue. The steepening of both curves in 2022 reflects the dominant contribution of the energy crisis year ($\approx$ EUR 7.9M combined). Post-2022 normalisation leads to flatter accumulation, though 2025 still shows materially higher combined revenue than 2023–2024, underscoring sensitivity to the prevailing price environment.

4.1.3 Data Processing Pipeline

The Day-Ahead spot price data underwent a systematic preprocessing pipeline to ensure consistency and economic interpretability. Raw prices at the **native ENTSO-E publication resolution** (predominantly 1h before 1 October 2025 and including 15-minute MTUs thereafter) for Swedish bidding zones SE3 and SE4 were retrieved from the ENTSO-E Transparency Platform via the `entsoe-py` Python client, covering 1 January 2020 (inclusive) to 1 January 2026 (exclusive).

Timestamps provided in UTC were converted to Swedish local time (CET/CEST) using the `Europe/Stockholm` timezone, with duplicate hours during Daylight Saving Time transitions removed (keeping the last observation). Short gaps (≤ 2 consecutive hours) were handled via time-based linear interpolation (`method='time'`, `limit_direction='both'`); longer gaps were retained as missing to avoid introducing bias.

All nominal prices were adjusted for inflation to a common 2025 EUR basis using the Swedish all-items Harmonised Index of Consumer Prices (HICP) from Eurostat (dataset `prc_hicp_midx`, CP00, 2015=100).¹ For each observation, the monthly HICP index was applied to rescale nominal prices, ensuring comparability across the six-year horizon.

The processed dataset, comprising approximately 52,500 hourly-equivalent observations before October 2025 and 15-minute resolution thereafter (about 96 intervals per day), was stored in Apache Parquet format

¹If the Eurostat API retrieval fails, the replication code falls back to a bundled local HICP series to keep the pipeline reproducible.

for efficient downstream use. Days with incomplete native interval counts (e.g., DST transitions) were excluded from the daily optimisation. The exact count of excluded days is logged in the replication code (chapter4_spot_arbitrage/logs/failed_days.txt).

4.2 Balancing Market Revenues

4.2.1 Price Dynamics and Summary Statistics

Table 4.2 summarises total balancing revenues under the stacked virtual portfolio assumption (activation + capacity streams).

Table 4.2. Balancing revenues: summary statistics (2020–2025, 2025 EUR). *Balancing revenues* denote the stacked sum of the five modelled ENTSO-E streams (aFRR/mFRR activation; aFRR/mFRR/FCR capacity) under the virtual-portfolio assumption. *Combined is calculated from the daily sum of SE3 and SE4 balancing revenues; therefore, the combined median, standard deviation, minimum, and maximum are computed from the combined daily series and do not necessarily equal the sum of zone-level summary statistics. The combined series does not account for physical capacity conflicts in a single asset.

Metric	SE3 (EUR)	SE4 (EUR)	Combined* (EUR)
Mean daily profit	54,059	62,270	116,329
Std. dev. daily profit	54,223	65,193	116,241
Median daily profit	41,245	45,006	87,184
Min daily profit	0	0	0
Max daily profit	339,510	640,901	744,506
Total Revenue	118.55M	136.56M	255.11M

4.2.2 Empirical results and visual evidence

This subsection summarises balancing-market behaviour and the associated revenue patterns, supported by figures.

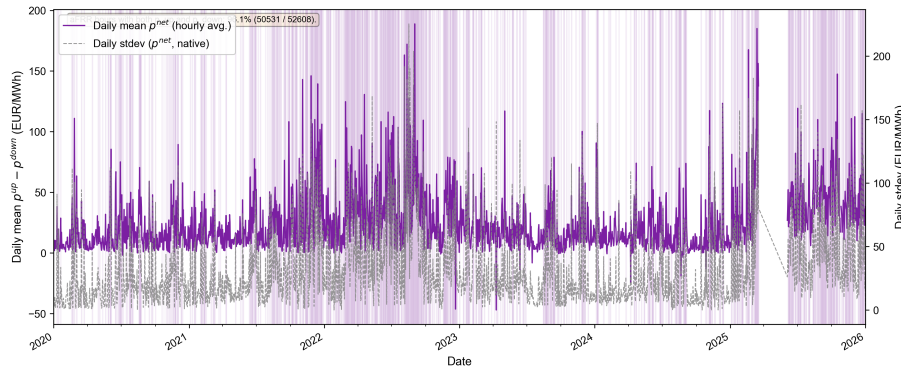


Fig. 4.6. aFRR activation prices (illustrative stream)

Figure 4.6 shows the net aFRR activation price signal ($\pi^{\text{net}} = \pi^{\uparrow} - \pi^{\downarrow}$) used as an illustrative example stream. For visual clarity the series is shown at hourly resolution, while optimisation in the replication package uses native MTUs. The signal exhibits higher volatility than Day-Ahead prices, with frequent crossings

of zero indicating shifts between up- and down-regulation demand.

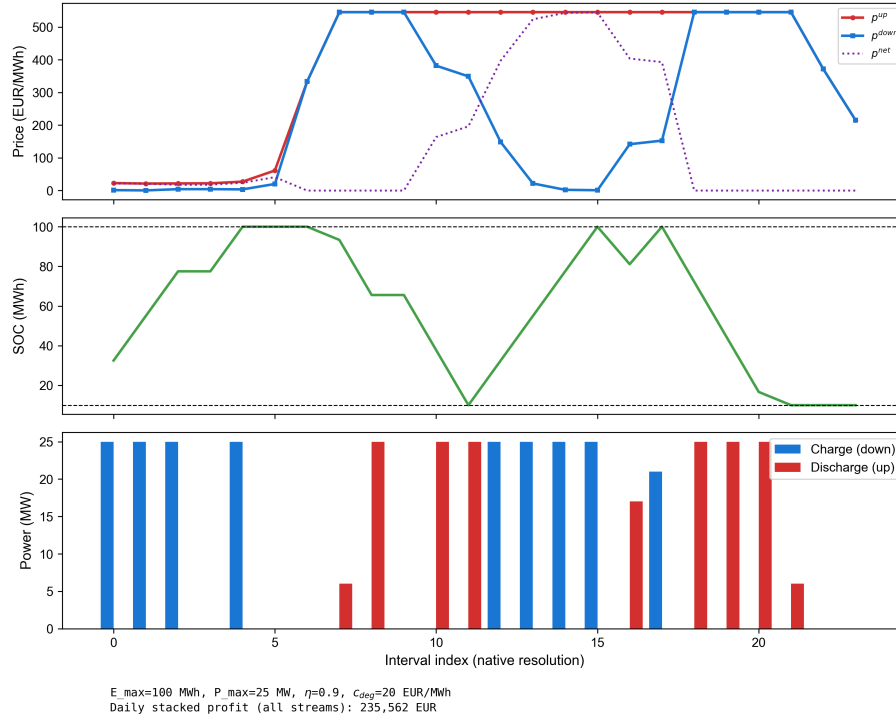


Fig. 4.7. aFRR activation dispatch (illustrative day)

Figure 4.7 illustrates an optimal stacked dispatch example on a high-volatility day. The battery responds to the net activation price signal, charging when $\pi^{\text{net}} < 0$ (down-regulation) and discharging when $\pi^{\text{net}} > 0$ (up-regulation). The resulting daily stacked profit (as reported in the figure footer for the selected day) highlights the upside potential of balancing settlement under perfect foresight and the virtual portfolio convention.

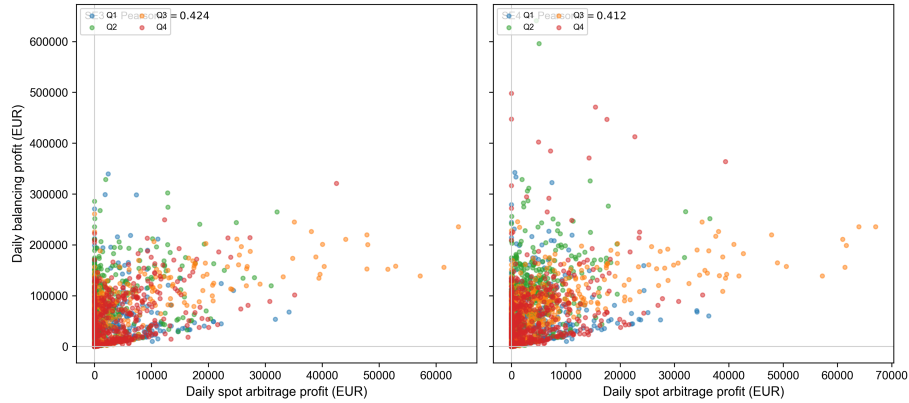


Fig. 4.8. Spot vs total balancing revenues

Figure 4.8 plots daily spot arbitrage profit against **total balancing** revenue BR_d (stacked streams). The figure indicates a moderate positive correlation (SE3: $r = 0.424$; SE4: $r = 0.412$), implying that high-profit days tend to occur simultaneously across the two revenue legs.

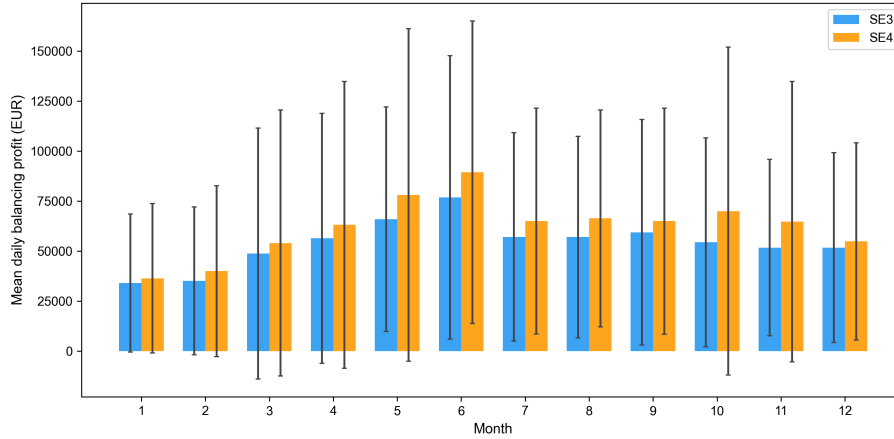


Fig. 4.9. Total balancing revenue seasonality

Figure 4.9 shows mean daily **total balancing** revenue by calendar month. The results indicate clear month-to-month variation, with relatively higher values around late spring and early summer, especially in May and June. This suggests that simulated balancing revenue is not driven only by winter demand conditions, but may also reflect product-specific reserve price dynamics, market stress events, and the timing of available capacity-price data. The large error bars also indicate strong within-month dispersion, so the monthly pattern should be interpreted cautiously.

4.2.3 Data Processing for Balancing and Reserve Prices

Activated balancing energy prices (document type A84; business types A96/aFRR and A97/mFRR; process type A16) and contracted reserve prices (document type A81; process types A51, A47, and A52 for aFRR, mFRR, and FCR capacity) were retrieved for SE3 and SE4. Series were aligned to Europe/Stockholm, clipped to $[-150, 500]$ EUR/MWh (activation and capacity price series), and deflated to 2025 EUR using Eurostat HICP (CP00). Optimisation used native MTUs; no pre-optimisation hourly averaging was applied.

Activated volume data (A83) are not available via the public API for Sweden; activation revenues are therefore price–dispatch upper bounds. Capacity revenues are computed only from 2022 (aFRR) and 2024 (mFRR, FCR) onward. Implementation details are in `chapter4_balancing/` (replication package).

4.3 Proxy Model: Spot Indicators and Total Balancing Revenues

The goal of the proxy model is to test whether observable spot dynamics can partly explain simulated *total balancing* revenue benchmarks.

4.3.1 Regularised linear proxy: coefficients and shrinkage

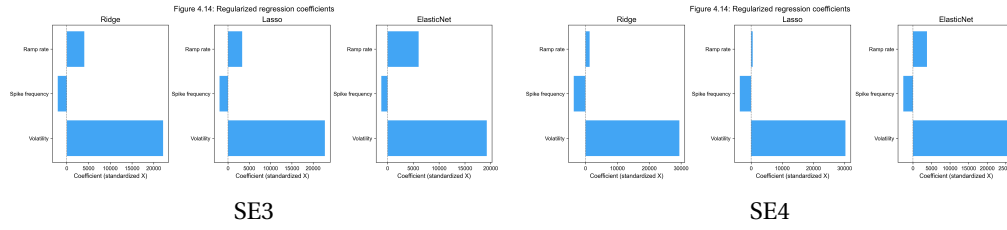


Fig. 4.10. Regularized coefficients for SE3 and SE4

Figure 4.10 compares coefficient vectors across Ridge, Lasso, and Elastic Net. All three shrinkage estimators yield similar sign patterns: volatility and ramp rate load positively, while spike frequency shows smaller or slightly negative contributions. This consistency across penalties suggests stable relationships rather than model-specific artefacts, supporting the use of regularised linear models as interpretable proxies.

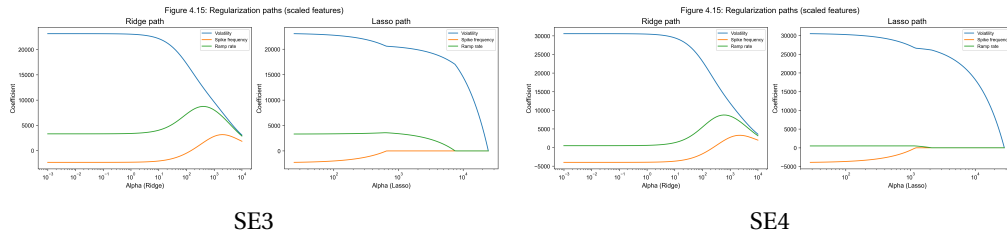


Fig. 4.11. Regularization paths for SE3 and SE4

Figure 4.11 shows how coefficients evolve as the penalty strength (α) varies. Volatility remains positive across the full path, while spike frequency and ramp exhibit greater sensitivity to λ , reflecting their higher collinearity with other indicators. These paths justify the choice of cross-validated α values that balance bias and variance.

4.3.2 Nonlinear benchmark and interpretability

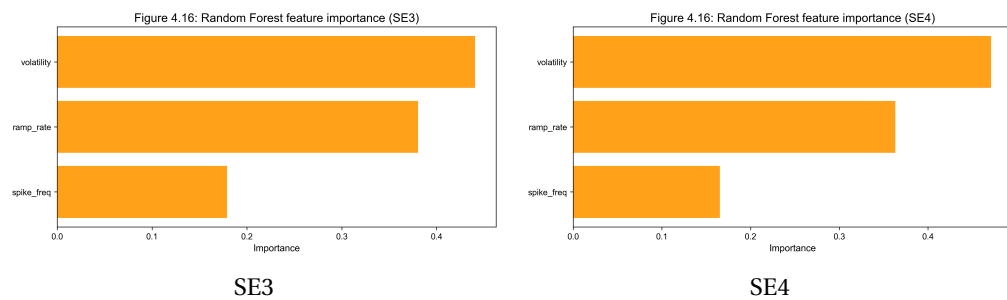


Fig. 4.12. Random forest feature importance for SE3 and SE4

Figure 4.12 shows Random Forest feature importance (impurity-based). **Volatility** contributes the largest share in both zones, followed by **ramp rate**, with **spike frequency** third (SE3: 0.441 / 0.380 / 0.179; SE4: 0.471 / 0.363 / 0.165). This ordering differs from linear coefficient magnitudes, reflecting trees' ability

to allocate importance across correlated predictors. The cross-zone stability of the ranking supports using spot indicators as coarse proxies, while the limited test R^2 cautions against over-interpreting magnitudes as causal elasticities.

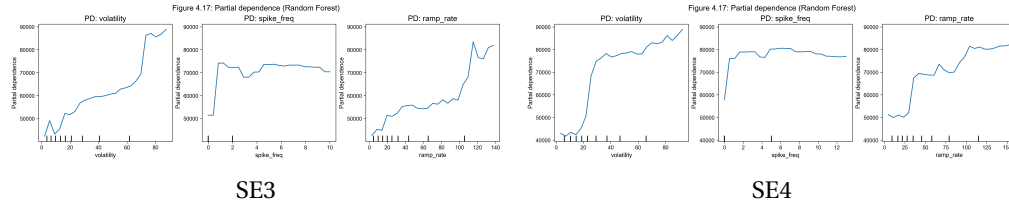


Fig. 4.13. Partial dependence plots for SE3 and SE4

Figure 4.13 illustrates marginal associations between each indicator and predicted balancing profit, holding other features averaged. The plots are broadly consistent with higher stress indicators mapping to higher predicted revenue on the training support, though local slopes can flatten; therefore they should be read as qualitative guidance rather than global linear elasticities. The near-linear patterns for volatility and ramp support the use of linear proxy specifications for scenario analysis.

4.3.3 Out-of-sample performance and diagnostics

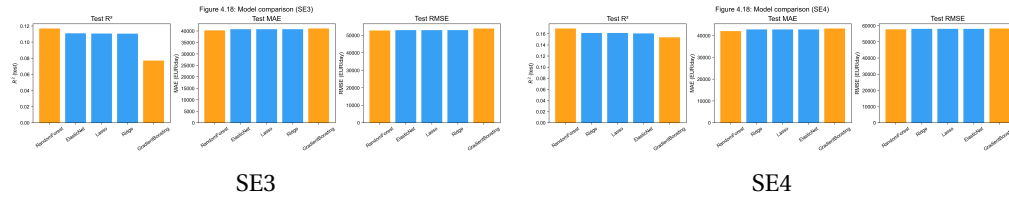


Fig. 4.14. Model comparison for SE3 and SE4

Figure 4.14 compares out-of-sample performance across models. Random Forest achieves the highest test R^2 for both zones (SE3: 0.117, SE4: 0.170), but overall explanatory power remains limited across all models. This motivates interpreting the proxy as a coarse screening tool rather than a high-accuracy forecasting model.

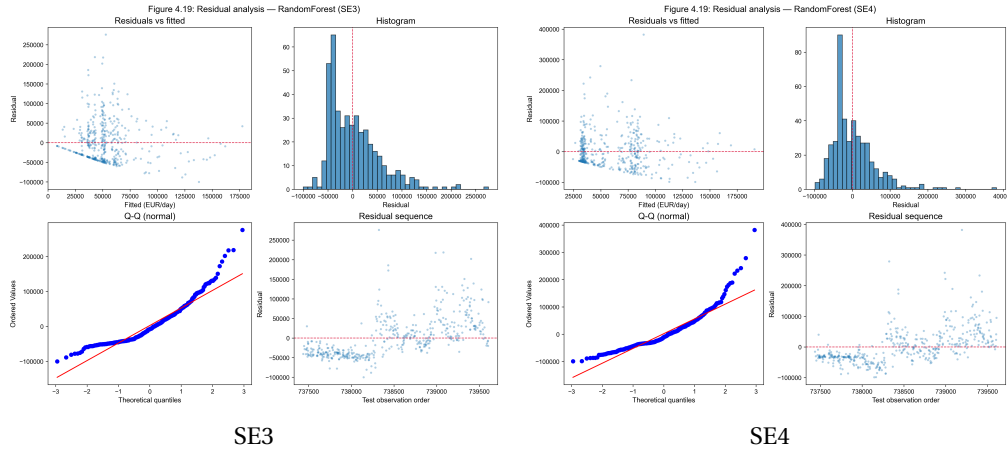


Fig. 4.15. Residual analysis for Random Forest model in SE3 and SE4

Figure 4.15 shows Random Forest residual diagnostics on the test set. The residuals vs. fitted plot indicates heteroscedasticity and heavy tails, which are typical for daily power revenues and motivate reporting error metrics alongside R^2 rather than relying on strict Gaussian error assumptions.

Hold-out performance summary

Five models were compared using a 20% hold-out test set. As shown in Table 4.3, **Random Forest** achieves the highest test R^2 for both zones, though differences to regularised linear models are modest.

Table 4.3. Proxy models: hold-out performance (20% test set; EUR/day)

Model	Type	SE3		SE4	
		R^2_{test}	MAE / RMSE	R^2_{test}	MAE / RMSE
Random Forest	Tree-based	0.1169	40,259 / 52,700	0.1696	42,036 / 57,704
Elastic Net	Regularised	0.1109	40,784 / 52,879	0.1608	42,796 / 58,011
Lasso	Regularised	0.1105	40,771 / 52,890	0.1615	42,789 / 57,985
Ridge	Regularised	0.1104	40,784 / 52,891	0.1616	42,795 / 57,983
Grad. Boosting	Tree-based	0.0770	41,092 / 53,877	0.1538	43,249 / 58,252

4.3.4 Data Preparation for Proxy Modeling

The proxy model was calibrated using a daily panel constructed by merging processed Day-Ahead spot prices (Section 4.1) with simulated **balancing revenues** from the deterministic optimisation (Section 4.2). For each calendar day in Swedish local time (Europe/Stockholm), three spot indicators were computed from the processed_prices output used for regression: if the underlying Day-Ahead index step is **shorter than 59 minutes** (15-min MTUs), the series is first resampled to **1h mean** for indicator construction, while the dispatch optimisation continues to use the native-resolution series. Indicators were then computed from the hourly series: volatility (standard deviation of 24 values), spike frequency (count of hours exceeding the pooled sample 95th percentile),

and ramp rate (maximum absolute hour-to-hour change). The intraday spread (max–min) was computed for diagnostic purposes only and excluded from estimation due to near-perfect collinearity with volatility ($r \approx 0.98$).

Daily observations with more than two missing hourly prices were discarded to ensure indicator reliability; additionally, a calendar day was retained only if it contained at least **approximately 22 valid hourly points** after resampling. The merged panel was randomly split into training (80%) and test (20%) sets with a fixed seed for reproducibility. Features were standardised (zero mean, unit variance) using parameters fitted on the training set only, then applied to both splits for Ridge, Lasso, and Elastic Net estimation; tree-based models used raw features. Multicollinearity was assessed via Variance Inflation Factors (VIF); after removing the spread diagnostic, volatility VIF falls to 13.76 (SE3) and 10.78 (SE4), while ramp-rate VIF remains elevated (11.69 in SE3; 9.41 in SE4).

The target variable—daily **balancing** profit BR_d —was simulated using the virtual-portfolio optimisation described in Section 4.2, not observed metered cash flows. Consequently, the proxy relationship should be interpreted as an associational mapping for scenario screening, not a causal identification of balancing price drivers. Full implementation details, including hyperparameter tuning via 5-fold cross-validation and diagnostic outputs, are documented in the replication code (`chapter4_regression/`).

4.4 Scenario Analysis: Revenue Sensitivity

The calibrated Lasso proxy relationship was applied to a reproduced historical baseline and three stressed price scenarios (four price paths in total). For scenario application, the Lasso proxy relationship was refit on the full historical panel (with 5-fold cross-validation for λ selection) rather than the 80/20 split used for hold-out evaluation. **Note on baseline totals:** in this section, the baseline is a *proxy-reproduced* baseline rather than the direct historical optimisation total reported in Sections 4.1 and 4.2. Balancing-revenue values in the scenario module are mechanically inferred through the proxy relationship, while balancing price paths are not directly shifted. Therefore, the baseline totals in this section should be interpreted as the reference level for scenario comparison, not as a reproduction of the historical optimisation benchmark. The difference between the proxy-reproduced baseline and the earlier optimisation totals reflects proxy fitting error and re-fitting on the full historical panel. In addition, applying the proxy outside the historical price regime (especially under the High Renewable intraday shape) constitutes extrapolation and should be interpreted cautiously.

For the scenario analysis, the total scenario-level upper-bound revenue

estimate is defined as:

$$TR_{d,z}^{\text{scen}} = AR_{d,z}^{\text{scen}} + \widehat{BR}_{d,z}^{\text{scen}} \quad (4.1)$$

where $TR_{d,z}^{\text{scen}}$ is the total scenario-level upper-bound revenue estimate for day d in bidding zone z under a given scenario, $AR_{d,z}^{\text{scen}}$ is the Day-Ahead arbitrage revenue calculated from the scenario spot price path, and $\widehat{BR}_{d,z}^{\text{scen}}$ is the proxy-inferred balancing revenue component inferred from the scenario spot indicators. The hat symbol shows that balancing revenue is not directly optimised from balancing-market prices, but mechanically inferred through the proxy relationship.

For comparison, the historical optimisation benchmark is defined as:

$$TR_{d,z}^{\text{hist}} = AR_{d,z}^{\text{hist}} + BR_{d,z}^{\text{hist}} \quad (4.2)$$

where $AR_{d,z}^{\text{hist}}$ is the historical Day-Ahead arbitrage revenue and $BR_{d,z}^{\text{hist}}$ is the historical simulated balancing revenue calculated from the available balancing and reserve price series. Therefore, Table 4.4 compares the cumulative historical optimisation benchmark, $\sum_d TR_{d,z}^{\text{hist}}$, with the cumulative proxy-reproduced scenario baseline, $\sum_d TR_{d,z}^{\text{scen}}$.

Table 4.4. Cumulative historical optimisation benchmark and proxy-reproduced scenario baseline by bidding zone, 2020–2025, real 2025 EUR.

Zone	Historical optimisation (spot + balancing)	Proxy baseline (scenario)	Difference
SE3	125.10M	195.55M	+70.45M
SE4	145.07M	247.70M	+102.63M

The proxy-reproduced scenario baseline is used only as the internal comparison benchmark for scenario analysis. It should not be read as a replacement for the historical optimisation results in Sections 4.1 and 4.2, because its balancing component, $\widehat{BR}_{d,z}^{\text{scen}}$, is inferred from spot indicators rather than directly computed from historical balancing and reserve price series.

4.4.1 Scenario design and price profiles

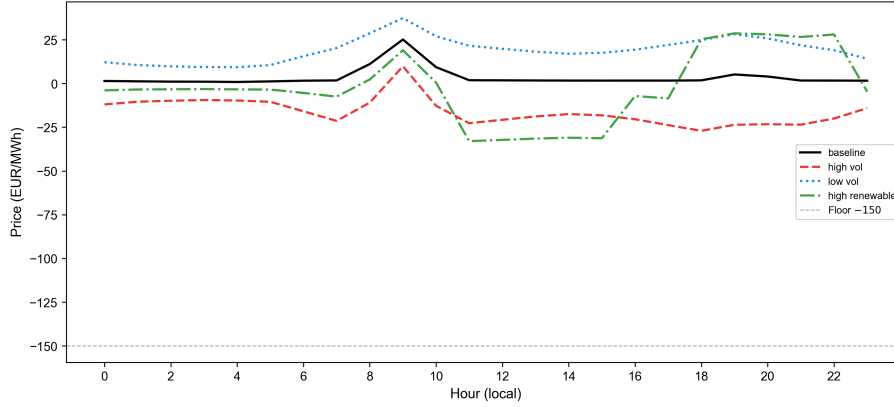


Fig. 4.16. Scenario price profiles

Figure 4.16 illustrates the construction of scenario price profiles on a representative day. High/Low Volatility scenarios amplify or dampen stochastic deviations around the same structural mean (μ_t). The High Renewable scenario introduces explicit intraday shifts: midday suppression (-25 EUR/MWh, 11–15h) and evening amplification (+35 EUR/MWh, 18–22h), capturing the stylised impact of high solar penetration on price formation.

4.4.2 Revenue uplift across scenarios

In Figure 4.17, annual revenue refers to the scenario-level upper-bound revenue estimate, including Day-Ahead arbitrage revenue and the proxy-inferred balancing component.

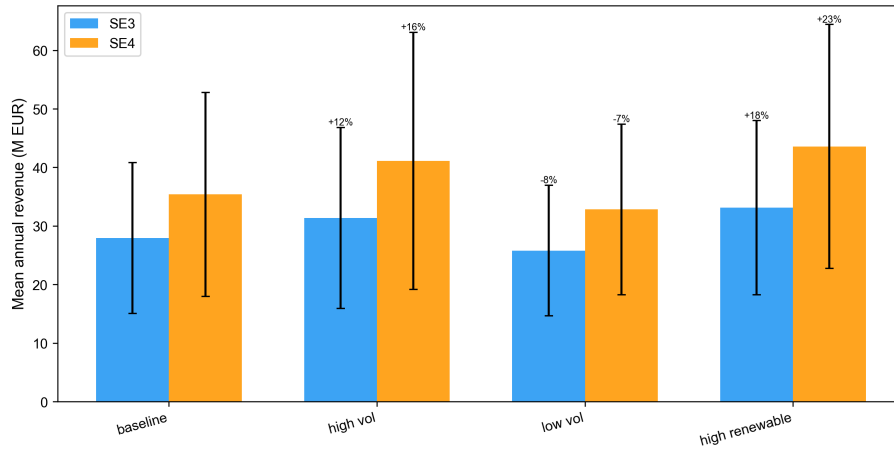


Fig. 4.17. Mean annual scenario-level upper-bound revenue estimate across scenarios.

Figure 4.17 compares mean annual scenario-level upper-bound revenue estimates across the baseline, High Volatility, Low Volatility, and High Renewable scenarios. Relative to the baseline, High Volatility increases total revenues by +12.3% (SE3) and +16.1% (SE4), while Low Volatility reduces them by -7.6% (SE3) and -7.3% (SE4). The High Renewable scenario yields the largest uplift

(+18.5% in SE3; +23.2% in SE4), reflecting the combined effect of structural intraday shifts and moderately higher volatility. Error bars show inter-annual dispersion.

4.4.3 Revenue composition and cumulative paths

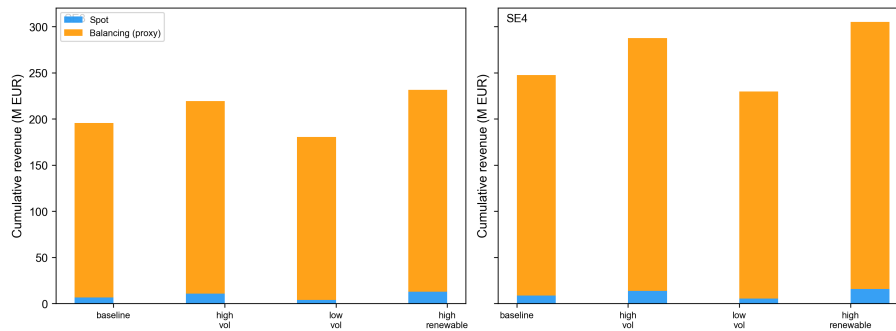


Fig. 4.18. Revenue composition

Figure 4.18 decomposes cumulative revenue into spot arbitrage (blue) and proxy-inferred balancing (orange) components. Proxy-inferred balancing components dominate in all scenarios (approximately 94–97% of total), underscoring that while spot indicators are the *input*, the scenario results are highly dependent on the proxy relationship and should be interpreted as screening-level sensitivity outputs. This dominance should be interpreted within the upper-bound virtual-portfolio framework: it does not imply that a single physical battery could directly realise the same revenue composition after accounting for capacity conflicts, SOC management, bidding restrictions, and activation-volume uncertainty. As volatility rises, the spot share increases modestly, but the proxy-inferred balancing component remains dominant.

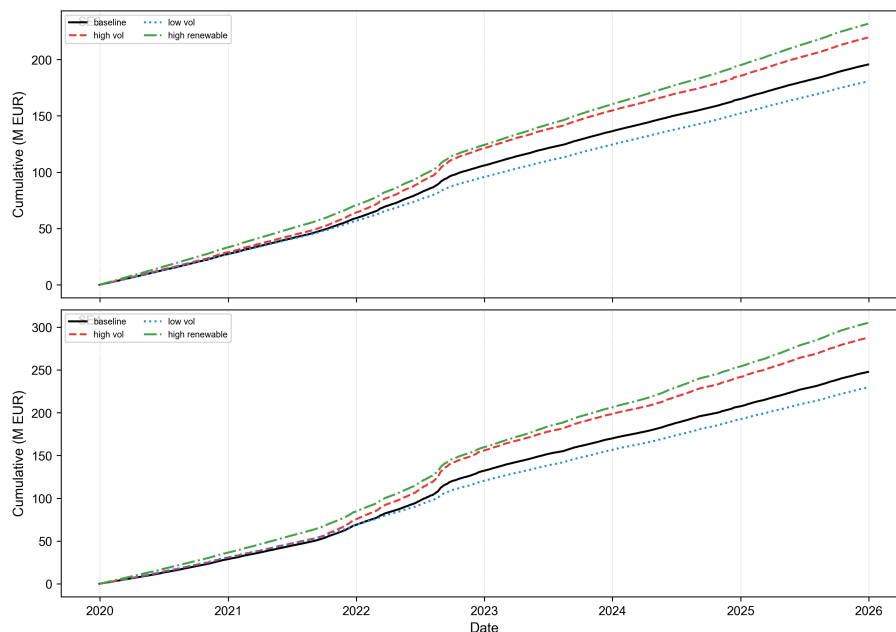


Fig. 4.19. Cumulative revenue paths

Figure 4.19 shows cumulative revenue paths for all scenarios. Divergence between scenarios accelerates after 2021, with High Renewable pulling ahead most strongly. The 2022 crisis year remains a dominant contributor across all scenarios, highlighting the path-dependence of cumulative returns and the importance of stress-testing investments against multiple futures.

4.4.4 Sensitivity analysis and key drivers

To make the sensitivity analysis clearer, Table 4.5 summarises the one-at-a-time parameter changes used in Figure 4.20.

Table 4.5. One-at-a-time sensitivity cases and parameter changes.

Sensitivity case	Description
$k_{HV} = 1.5$	Increases spot price volatility by scaling stochastic price deviations by 1.5.
$k_{LV} = 0.6$	Reduces spot price volatility by scaling stochastic price deviations by 0.6.
$k_{RE} = 1.2$ only	Applies the renewable scenario volatility scaling only, without changing midday or evening prices.
Midday (−25) EUR/MWh only	Reduces midday prices to represent stronger solar generation effects.
Evening (+35) EUR/MWh only	Increases evening prices to represent higher evening scarcity or peak demand.

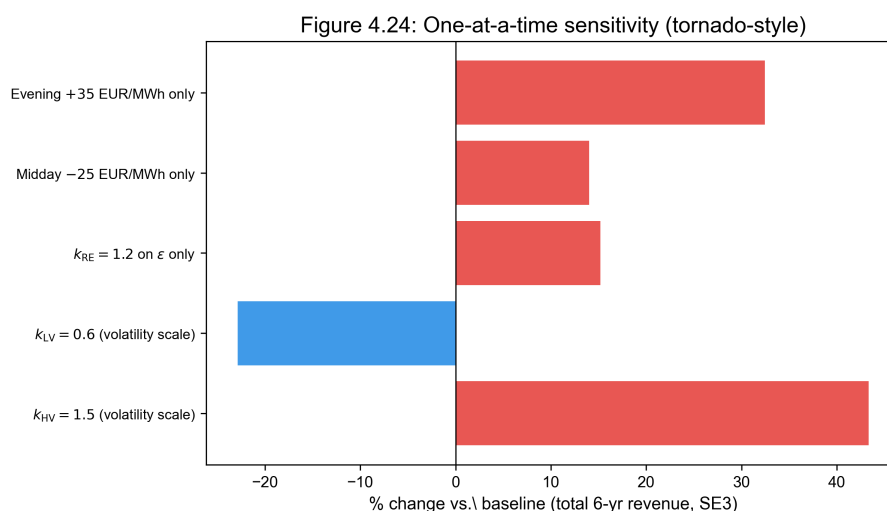


Fig. 4.20. Sensitivity tornado diagram (one-at-a-time parameter changes; the High Renewable scenario combines multiple parameter changes simultaneously)

Figure 4.20 reports the one-at-a-time sensitivity results based on the parameter changes described in Table 4.5. The largest isolated positive impact comes from the High Volatility scaling case ($k_{HV} = 1.5$), followed by evening price amplification (+35 EUR/MWh). The Low Volatility case ($k_{LV} = 0.6$) produces a negative impact relative to the baseline. These results suggest that stochastic volatility changes are an important driver of scenario-level revenue sensitivity,

while evening price peaks also contribute strongly. The full High Renewable scenario combines several effects simultaneously, explaining its larger total uplift in the scenario comparison. Importantly, the High Renewable scenario gives the largest overall uplift because it combines multiple changes, while the tornado diagram isolates drivers one at a time—these two results therefore answer slightly different questions.

Table 4.6. Scenario analysis: estimated total revenue (6-year cumulative). These totals are proxy-based scenario benchmarks and should be interpreted primarily in relative terms.

Scenario	SE3 Total (EUR)	SE4 Total (EUR)
Baseline	195.55M	247.70M
High Volatility	219.53M (+12.3%)	287.65M (+16.1%)
Low Volatility	180.59M (−7.6%)	229.70M (−7.3%)
High Renewable	231.69M (+18.5%)	305.06M (+23.2%)

Key takeaway

Proxy-inferred balancing components dominate in all scenarios, accounting for approximately 94%–97% of total income. This underscores that while spot market indicators are the *input*, the scenario results are highly dependent on the proxy relationship and should be interpreted as screening-level sensitivity outputs.

4.4.5 Scenario Data Construction Pipeline

Stylised spot price scenarios were constructed by decomposing the baseline hourly Day-Ahead series (Section 4.1) into a structural component μ_t (month–hour sample mean) and a stochastic residual $\varepsilon_t = p_t - \mu_t$. Three counterfactual scenarios were then generated: (i) High Volatility ($p_t^{\text{HV}} = \mu_t + 1.5\varepsilon_t$), (ii) Low Volatility ($p_t^{\text{LV}} = \mu_t + 0.6\varepsilon_t$), and (iii) High Renewable Penetration ($p_t^{\text{RE}} = \mu_t + 1.2\varepsilon_t$, with additional intraday shifts of −25 EUR/MWh for hours 11–15 and +35 EUR/MWh for hours 18–22). All trajectories were floored at −150 EUR/MWh to avoid economically implausible extremes.

For each scenario, daily spot indicators (volatility, spike frequency, ramp rate) were recomputed using the same rules as Section 4.3, with spike counts defined relative to the historical pooled 95th percentile to ensure cross-scenario comparability. The calibrated Lasso proxy relationship (refit on the full historical panel) was then applied to infer balancing-revenue components, which were aggregated with spot arbitrage profits from the deterministic optimisation to obtain scenario-level upper-bound revenue estimates.

This pipeline is a deterministic sensitivity exercise, not a forecast: it evaluates how scenario-level upper-bound revenue estimates respond to stylised structural changes in the spot price process.

4.5 Chapter Summary

The empirical results illustrate the behaviour of the proposed framework under public-data constraints. Proxy models provide only limited explanatory power for daily total balancing revenues (R^2_{test} up to 0.170), but scenario analysis still suggests, within the proxy-based sensitivity setting, that structural intraday shifts (High Renewable) yield the strongest uplift (+18.5% in SE3; +23.2% in SE4) relative to the reproduced baseline.

5

Discussion

This chapter interprets the empirical results presented in Chapter 4, contextualises them within the existing literature reviewed in Chapter 2, and discusses their implications for theory, practice, and future research. The discussion is structured around four themes: (1) interpretation of key findings, (2) methodological contributions and limitations, (3) practical implications for investment decision-making, and (4) directions for future research.

5.1 Conclusion

This thesis set out to examine to what extent a proxy-based framework using Day-Ahead spot market indicators can *explain* variation in long-term BESS revenue potential, including balancing-market components, under public-data constraints. The results suggest that such a framework can support transparent early-stage screening and sensitivity analysis, but it should not be interpreted as a reliable forecasting model for investable cash flows. By combining deterministic Day-Ahead arbitrage optimisation with spot-based indicators such as price volatility, spike frequency, and ramp rate, the framework provides a structured way to test whether accessible spot dynamics contain useful information about upper-bound revenue patterns when detailed balancing-market data and forecasts are unavailable.

Empirically, Day-Ahead spot indicators explain only a modest fraction of daily variation in simulated balancing and reserve-related revenue benchmarks. Therefore, the proxy is most useful for comparing zones, identifying stress-period revenue patterns, and running scenario-based sensitivity exercises that illustrate how total upper-bound revenue estimates respond under stylised spot price regimes. It is less suitable for predicting exact daily balancing revenues or for standalone investment decisions. In this sense, the framework is valuable as an interpretable empirical test and screening device, while its limitations are made explicit by the weak out-of-sample fit of the proxy models.

The thesis further indicates that, within the simulated upper-bound setting, the largest revenue contributions are concentrated in stress periods and are primarily linked to balancing and reserve-related flexibility services rather than Day-Ahead arbitrage alone. These results must be interpreted carefully. Perfect-foresight dispatch, the stacked virtual-portfolio assumption, product-level data limitations, and the absence of activation-volume information mean that actual operational revenues would likely be lower and more constrained.

This is consistent with the thesis design, where the virtual portfolio is used as an upper-bound benchmark rather than a feasible single-battery dispatch model.

The main practical implication is therefore twofold. First, investors and planners can use transparent spot indicators and scenario-based stress tests to support cautious early-stage comparison of BESS opportunities under uncertainty. Second, before any real investment decision is made, the upper-bound results should be followed by asset-specific modelling that includes bidding behaviour, activation probability, reserve volumes, degradation, market participation rules, and physical capacity allocation across services.

Overall, the contribution of this thesis is not the construction of a high-performing predictive model, but the empirical demonstration that publicly available Day-Ahead spot indicators have limited yet non-zero explanatory value for simulated balancing revenue benchmarks. Under data constraints, this provides a transparent first-step reference point, while also clarifying where more detailed modelling and proprietary data are required.

5.2 Interpretation of Key Findings

5.2.1 Spot Market Arbitrage: Upper Bounds and Path Dependence

The Day-Ahead arbitrage results show why spot arbitrage alone is not sufficient for long-term BESS revenue evaluation. Although spot prices are the most accessible public data source, the resulting arbitrage revenues are uneven and depend strongly on high-price-spread periods. This means that using historical average spot arbitrage revenue alone may give an incomplete view of future revenue potential. For this thesis, the main relevance of the spot arbitrage results is that they provide a transparent baseline revenue component and generate spot-based indicators for the proxy framework.

5.2.2 Balancing Revenues: Product Decomposition and Premiums

The balancing revenue results show why balancing and reserve-related components matter for long-term BESS revenue *assessment*. In this thesis, these components are not modelled as feasible revenue stacking for one physical battery. Instead, they are estimated in parallel under the virtual portfolio assumption and used as an upper-bound benchmark. The main relevance of these results is therefore not to claim directly achievable operational revenues, but to construct a historical balancing revenue benchmark against which the spot-indicator proxy can be tested and evaluated.

5.2.3 Proxy Model: Evidence on the Spot–Balancing Relationship

The regression analysis (Section 4.3) shows limited out-of-sample explanatory power for daily simulated *balancing revenues*, with Random Forest achiev-

ing the highest hold-out fit ($R^2_{\text{test}} = 0.1169$ in SE3; 0.1696 in SE4; Table 4.3). This supports using the proxy primarily for transparent scenario screening rather than high-accuracy forecasting.

Key observations:

- **Coefficient stability:** All three shrinkage estimators yield similar sign patterns (Figure 4.10), with volatility and ramp rate loading positively. This consistency across penalties suggests robust relationships rather than model-specific artefacts.
- **Multicollinearity management:** Removing the highly correlated Spread indicator ($r \approx 0.98$ with Volatility) and applying regularisation mitigates unstable coefficient estimates; diagnostics reported in Section 4.3.4 show that removing spread substantially reduces VIF for volatility, although residual correlation between volatility and ramp remains.
- **Tree-based robustness:** Random Forest feature importance places **volatility** first and **ramp rate** second in both zones, with **spike frequency** third (Figure 4.12), while partial dependence plots suggest predominantly weakly positive marginal associations on the training support (Figure 4.13). Nonlinearities therefore appear present in-sample but do not translate into large gains in out-of-sample R^2 on this split.

The residual diagnostics (Figure 4.15) reveal heteroscedasticity and heavy tails—typical for daily power revenues [20]. These patterns support reporting error metrics (MAE, RMSE) alongside R^2 rather than relying on strict Gaussian error assumptions.

5.2.4 Scenario Analysis: Sensitivity to Market Structure

Applying the calibrated proxy to stylised spot price scenarios as a sensitivity exercise (Section 4.4) suggests that total upper-bound revenue estimates are highly sensitive to volatility regimes and intraday structural shifts:

- **High Volatility:** +12.3% (SE3) and +16.1% (SE4) vs. baseline.
- **Low Volatility:** −7.6% (SE3) and −7.3% (SE4) vs. baseline.
- **High Renewable:** +18.5% (SE3) and +23.2% (SE4) vs. baseline (largest uplift).

The finding that the High Renewable scenario yields the largest revenue increase is noteworthy. It suggests that *structural intraday shifts* (e.g., solar-driven price shapes) may create more valuable arbitrage opportunities than pure volatility scaling alone. This aligns with analyses of how variable renewables affect balancing needs and price formation [12].

However, this result should be interpreted with caution: the proxy model was calibrated on historical data with limited exposure to high-renewable price structures. Extrapolating the spot–balancing relationship to structurally different regimes assumes stability that may not hold under rapid energy transition [1].

5.3 Methodological Contributions and Limitations

5.3.1 Contributions

This thesis makes three methodological contributions to the literature on BESS revenue estimation:

1. **Transparent proxy framework:** By combining native-resolution upper-bound dispatch with a spot-indicator proxy for *total* balancing revenues, this study provides a data-efficient screening tool under public-data constraints. Hold-out R^2 remains modest (up to about 0.12–0.17), which is informative: it signals that easily observed Day-Ahead shapes alone do not tightly pin down stacked balancing upper bounds, motivating conservative use in valuation rather than high-accuracy forecasting.
2. **Regularisation for multicollinearity:** The explicit handling of severe collinearity among spot indicators (via variable selection and Ridge/Lasso penalties) offers a reproducible protocol for energy econometrics applications where market variables are inherently correlated.
3. **Scenario-based sensitivity analysis:** Rather than point forecasting, the thesis applies the proxy to stylised price scenarios, enabling investors to stress-test projects against plausible market developments without requiring proprietary forward-looking balancing forecasts.

5.3.2 Limitations

Several limitations must be acknowledged:

- **Perfect foresight assumption:** The deterministic optimisation models assume known prices within each delivery day at native resolution (T_d intervals), yielding theoretical upper-bound revenues. Real-world trading with forecast errors would capture a lower proportion of these profits [3, 27].
- **Virtual portfolio assumption:** Revenues from spot arbitrage and balancing markets are estimated independently and aggregated post-hoc. A single physical battery faces state-of-charge competition and simultaneous service conflicts that may reduce achievable combined revenues [24].

- **Proxy relationship stability:** The calibrated relationship between spot indicators and balancing revenues assumes structural stability over time. Regulatory changes, market design reforms, or shifts in generation mix could alter this relationship, limiting out-of-sample applicability.
- **Heterogeneous MTUs and product windows:** Combined balancing totals mix hourly and 15-minute upper bounds and include capacity streams only from product-specific start years (2022/2024). Interpretation as a single homogeneous series requires caution.
- **Parameter point estimates:** Technical parameters (efficiency, degradation cost, cycle limits) are fixed at representative values. Sensitivity analysis could quantify revenue uncertainty under technical variation, but is beyond the scope of this study.

Despite these limitations, the methodology prioritises transparency and long-term usability over short-term forecasting precision. The theoretical upper-bound estimates provide a consistent benchmark for comparing market scenarios, while investors are advised to apply appropriate risk discounts for operational constraints.

5.4 Directions for Future Research

Several avenues for future work emerge from this study:

5.4.1 Methodological Extensions

- **Time-series validation:** Replace random train/test splits with time-ordered cross-validation to better respect serial dependence in daily revenue data.
- **Nonlinear proxy specifications:** Explore generalized additive models (GAMs) or gradient boosting to capture potential nonlinearities or interactions among spot indicators.
- **Multi-market co-optimisation:** Develop a unified optimisation model that explicitly accounts for physical capacity conflicts when a single battery participates in both spot and balancing markets.

5.4.2 Data and Scope Extensions

- **Expanded geographical scope:** Apply the framework to other European bidding zones (e.g., DE/LU, FR, UK) to test the generalisability of spot-balancing relationships.
- **Longer historical horizon:** Extend the calibration period beyond 2025 as more high-renewable data becomes available, enabling validation of proxy stability under structural change.

- **Incorporating activated volumes:** If Document Type A83 data becomes accessible, replace arithmetic averaging with volume-weighted aggregation for more accurate balancing price representation.

5.4.3 Integration with Broader Energy System Models

- **Coupling with capacity expansion models:** Embed the proxy-based revenue estimator within long-term energy system planning tools to assess how storage deployment responds to evolving market structures.
- **Policy scenario analysis:** Evaluate how specific policy interventions (e.g., capacity mechanisms, carbon price floors) affect estimated storage revenues through their impact on spot price dynamics.

5.5 Chapter Summary

This discussion has interpreted the empirical results through the lens of existing literature, highlighted methodological contributions and limitations, and outlined practical implications for investment decision-making. The key takeaways are:

1. Spot arbitrage revenues are highly path-dependent, with extreme events driving disproportionate value—a finding consistent with electricity market microstructure theory.
2. Balancing revenues exhibit moderate positive co-movement with spot arbitrage, implying that high-revenue days may occur simultaneously across the two revenue legs, while diversification benefits remain limited under the virtual-portfolio setting.
3. **Proxy evidence under modest fit:** Spot indicators deliver only modest hold-out R^2 for daily simulated balancing revenues (up to about 0.12–0.17; Table 4.3), yet regularised specifications remain interpretable and Random Forest provides a nonlinear benchmark. Together, this supports using the framework for transparent *screening* rather than high-accuracy forecasting of balancing cash flows.
4. Scenario analysis reveals high sensitivity to volatility regimes and renewable-driven price structures, with the High Renewable scenario yielding the largest revenue uplift.
5. The framework provides a practical, data-efficient tool for early-stage project evaluation, though investors should apply risk discounts to account for upper-bound assumptions.

Collectively, these findings support the conclusion that a proxy-based approach can be used to *test* and illustrate how total upper-bound revenue estimates respond to Day-Ahead spot dynamics under data constraints, offering

a transparent complement to more detailed stochastic models for long-term investment analysis.

AI Tools Discussions

In accordance with institutional academic integrity guidelines, the author discloses the limited use of AI-assisted tools during this research. These tools served strictly as auxiliary aids; all research design, analytical decisions, and scholarly interpretation were conducted independently.

- **Literature Screening:** Google Search and AI summarisation features were used to identify relevant publications and extract preliminary insights from abstracts. All sources were read in full and critically synthesised by the author.
- **Code Development:** Cursor (AI-assisted IDE) was employed to accelerate code scaffolding, debug scripts, and suggest syntax improvements for the optimisation and regression pipelines. All mathematical formulations, data logic, and validation steps were designed and verified manually.
- **Formatting & Typesetting:** AI-assisted formatting features within Overleaf were utilised to adjust LaTeX layout, standardise citation styles, and optimise figure/table placement. The academic content, structure, and argumentation remain entirely author-generated.

AI tools were not used to generate hypotheses, draft analytical sections, or produce final academic prose. The author retains full intellectual responsibility for the methodology, results, and conclusions presented in this thesis.

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List of Included Libraries and GitHub Codebase

The replication code covers: `chapter4_spot_arbitrage` (Day-Ahead arbitrage); `chapter4_balancing` (aFRR/mFRR activation and aFRR/mFRR/FCR capacity upper bounds); `chapter4_regression` (proxy on total balancing revenues); and `chapter4_scenario_analysis` (Day-Ahead scenarios + proxy balancing leg).

Code repository: https://github.com/jilintao/BESS_Thesis/tree/main

Python libraries (pinned versions)

The following packages are used by the replication pipelines. Versions are pinned for reproducibility.

```
entsoe-py==0.7.11
numpy==2.1.3
pandas==2.2.3
scipy==1.14.1
scikit-learn==1.5.2
statsmodels==0.14.4
pulp==3.3.0
matplotlib==3.8.3
seaborn==0.13.2
requests==2.32.3
pyarrow==17.0.0
```

Solver

Daily dispatch optimisation is implemented in PuLP and executed using the bundled COIN-OR CBC solver (`PULP_CBC_CMD`). If CBC is unavailable, alternative solvers (e.g., GLPK) can be configured in PuLP.